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Science Advocacy in Danger

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Abstract

Systemic changes will take place by creating conditions for effective advocacy and promotion of science and scientists through formation of relevant public and non-governmental institutions, restructuring of existing communication platforms, and professional development of science activists. Science requires collective advocacy and, accordingly, work with stakeholders, i.e., government, business, civil society, and international partners.

Laws define the rights of scientists and researchers, but it is not easy to exercise these rights. For the most part, scientists are not aware of the tools to protect their rights and do not always recognize needs as problems to be solved. There is a need to create outreach or educational programs to raise scientists' awareness on this topic.

Cross-sectoral partnership, social cohesion, and creation of communication platforms facilitate effective interaction. A well-established model of cooperation between business and scientists in the frame of grant funding, where scientists can be seen as customers, can become a platform for advocating for preservation, development, and popularization of science at various levels. Due to the constant changes in the context, effective advocacy requires regular monitoring of the mechanisms of ensuring scientific activity, and development of communication of science with stakeholders and civil society.

Based on the obtained data, a number of **recommendations** were developed

For scientists involved in science advocacy

1. **Diversification of scientific** agenda should be advocated to the media and media influencers. It is necessary to formulate principles of science communication and create a code of ethics for science communication. It is necessary to study the possibilities of social media related to advocacy campaigns, form the informational ground and advocacy goals, and involve influencers who are trusted by both scientists and society. The available media resources in higher education institutions (newspapers, radio, promotional platforms, etc.) should be monitored to find out their ability to publish materials about science in a clear language.
2. In **desk research**, the models and formats that facilitate interaction of science with stakeholders should be analyzed and outscaled to inform about science in interaction with journalists and other media representatives. The discussions

with leading media and bloggers working for the general public should be arranged. In particular, there is a need for translation and publication of scientific and applied literature on science advocacy.

3. **International advocacy** will help to increase the level of subjectivity of Ukrainian science and give foreign partners a clearer understanding of the current situation in Ukrainian science and its needs. The best way to advocate your interests is to meet international stakeholders in person. We recommend interested Ukrainian scientists and organizations to actively establish contacts with foreigners (through mutual acquaintances, scientific conferences or schools, participating in them online if it is impossible to travel, by attracting foreign scientists to scientific schools in Ukraine, participating in working groups with foreigners). Knowledge of English is an important condition for effective advocacy, so it is necessary to create conditions for learning it.
4. The level of self-awareness, organization and preparedness of the scientific community should increase. Through **networking**, i.e., creation of formal public associations and informal networks in the field of science, the intellectual potential of advocacy should be strengthened which will have a positive impact on the quality of submitted proposals. Public activity of scientists in social media should be combined with activity during public discussions which will increase the effectiveness of advocacy. As the tentative calendar of public discussions of the Ministry of Education and Science is approved and published before the beginning of each year, stakeholders can plan their advocacy activities for the year ahead.
5. The scientists who want positive changes in the development of Ukraine's scientific field should try to realize their potential in the **public sector**. In this case, according to the legislation of Ukraine, civil service can be combined with scientific activity.

For authorities and public institutions

1. The authorities should be more proactive in informing the scientific community about **public discussions** and explain the importance of a particular regulatory act. An elaborate communication system should be in place to attract more participants in the discussions and timely inform about events. To do this, it is necessary to use social networks and social media, not only official channels, but also to send newsletters to thematic groups and forums. Another way would be to create a Google group for public discussions with the possibility of open access for interested parties. It would be useful to create a database of contacts of experts who participated in previous public discussions and send them invitations to participate in the discussions. Moreover, the list of possible formats for public discussions should be expanded (with account of safety requirements) which is currently limited to digital consultations.
2. The practice of **discussing science development strategies** should be improved. First, it is necessary to develop and test a methodology for conducting strategic discussions with an emphasis on synchronous communication (online and offline) in small groups with the participation of a moderator/facilitator. Second, strategic events should be held regularly and on a more systematic basis, and the scientific community should be involved in discussing the conceptual foundations of public policy at the initial stage of development. The Science for Policy principles should be implemented in the scientific field. Public hearings should not be formal, but should serve to conceptualize, actualize or problematize the needs, and can be held both on the initiative of internal stakeholders and

[1] Futures of Science for Policy in Europe: Scenarios and Policy Implications. URL: *https://research-and-innovation.ec.europa.eu/news/all-research-and-innovation-news/futures-science-policy-europe-scenarios-and-policy-implications-2023-10-10_en*

international organizations, in cooperation with communities and NGOs.

3. The role of **the National Council for Science and Technology Development** in shaping state policy in the field of science should be strengthened given its importance as a key element of the science advocacy mechanism. For this purpose, it is important to implement the provision of the Law of Ukraine "On Scientific and Scientific and Technological Activities", according to which the National Council should meet at least four times a year. Given the specifics of martial law, which may make it impossible for the Prime Minister to attend its meetings as the head of the National Council, the meetings may be chaired by the chairmen of the Scientific and Administrative Committees in turn. The recommendation to hold meetings of the Administrative Committee on the eve of each meeting of the National Council, i.e., at least four times a year, is important. In order to improve the qualifications of newly elected members of the Scientific Committee who do not have experience in lawmaking, the Cabinet of Ministers, with the involvement of specialists from the Parliament (upon agreement), may develop and conduct intensive and long-term training courses on the specifics of lawmaking according to the legislation of Ukraine.
4. The existing **advisory bodies** should be further developed and, in cooperation with other NGOs, create conditions for establishment of new councils representing different target groups (councils of women scientists, council of regional scientists, council of basic sciences, etc.).
5. In order to facilitate more effective science advocacy, the list of **open data** on science which are mandatory for publication should be expanded. This will allow establishing more meaningful and constructive communication on the development of science.
6. **The National Academy of Sciences** should actively and systematically cooperate with the public sector on a partnership basis, including volunteer initiatives, charitable foundations, and NGOs, develop joint solutions, jointly implement projects to meet the needs of science, and conduct advocacy campaigns.

For public sector

1. **In order to strengthen the intellectual potential** of advocacy, it is necessary to develop workshops and trainings, both informational and specifically advocacy, in particular on public participation in lawmaking. These events should include real drafts of legal acts that are currently submitted or planned for discussion as practical exercises. To conduct courses in the context of limited public finances, funds from international partners should be attracted. To increase the culture of discussion, cross-sectoral events, trainings on facilitation, netiquette, and media communications should be organized. Cooperation with stakeholders can begin with joint trainings on science advocacy (with government, business, media) that will include brainstorming and development of project ideas.
2. **Regular public monitoring** of the work of institutions that are called to act in the interests of scientists (National Academy of Science of Ukraine (NASU), National Research Foundation of Ukraine (NRFU), National Council for Science and Technology, trade unions, etc.) should be ensured to determine the effectiveness and transparency of their work, and promote public awareness of the results.
3. Involvement of NGOs as partners in science advocacy, as well as for cooperation with international donors for community development. The projects to build advocacy teams should be developed. **Promoting networking** proves itself an effective tool for developing science advocacy. It was thanks to networking that during the first year of the full-scale war it became possible to organize processes to help scientists from

abroad, provide access to those scientists who need support, etc. Advocacy actions that will facilitate creation of networking spaces by institutions include organizing special events for interinstitutional, cross-sectoral, and cross-disciplinary interaction, supporting networking platforms, websites, and developing centers of communication and interaction. It is important to inform about the experience of implementing changes in other areas of society, applying existing models, but with account of the needs of both general public and a group for whose benefit advocacy is applied.

4. **Translation of foreign research** on science advocacy, recommendations, and manuals into Ukrainian would be helpful in implementing advocacy activities in Ukraine, especially since there are almost no studies on advocacy of Ukrainian science, although there are analytical reports and described advocacy practices in related areas (administration of relocated universities, some achievements of reforms in the field of medicine, social policy, education, examples of local policy changes, in particular in education, culture towards inclusion, lobbying and implementation of legislative changes on human rights protection, gender policy, combating discrimination and violence).

For higher education institutions

1. **The training of specialists who will facilitate science advocacy**, such as science managers, science journalists, science editors, etc. could be a separate area of work. Their responsibilities and competencies will include the ability to develop the content from key messages to narratives for dissemination to different audiences (stories about scientists who saved the world, interesting facts about Ukrainian science that will increase its visibility in the information field), and organize cross-sectoral discussions, direct action campaigns to actualize science in the context of values, motivate people to unite around the values of which science is a part, create spaces for communication about science, i.e., media platforms, platforms, physical spaces (temporary — exhibitions, presentations, and permanent ones) in science parks, libraries, universities, etc.
2. For capacity development of people who advocate for science (mediators, educators, promoters) **curricula** and lectures on communication, advocacy, and promotion of science should be developed for scientists in cooperation with stakeholders at the master's level; the topics and tasks on science coverage, promotion and advocacy should be integrated into the existing curricula and courses in the humanities at the bachelor level (media communications, journalism, cultural management, event management, project production, etc., and the programs to improve the skills of existing scientists should be implemented.

For mass media and media influencers

1. **Providing access to information on the results of scientific activities.** Raising public awareness of science through dissemination of balanced, reliable, and accessible content will increase the level of self-awareness, organization, and preparedness of the scientific community. One of the tasks of journalism is to inform people about the world (promptly, reliably, with the necessary completeness, with account of different views, etc.). Journalism, in particular mass media, should develop the space for informing and communicating with science — preparing news, communicating with experts, developing new formats — talk shows, popular science programs, reports from national parks, etc. Cooperation with media editors at both national and local levels can be effective in covering science in the media discourse. Interaction with regional media by providing them with information about science helps create spaces for communication and preparation of popularization content which will allow for effective advocacy of science to local

authorities, administration of scientific institutions and business, etc.
2. The social media monitoring results show that **regions** need to be better informed about their activities, otherwise science advocacy will be focused only on certain regions which will lead to uneven public support for science in the implementation of certain political and economic decisions.
3. An important goal of advocacy is **building trust** in the scientist in society as an active public figure who has weight and can have influence at the political level. To achieve this goal, it is important to show the lives of scientists as people with ordinary needs, not just scientific ones. The interest in historical dates and participation in cultural events, political discussions and celebrations emphasizes the civic engagement of scientists. Unfortunately, the format of awarding scientists in Ukraine is not yet used as an advocacy tool, like the Nobel Prize, for example. When covering the awards, the name of the prize and name of the nominee are usually reported, the life story of the scientist and their career are described, but not enough attention is paid to the essence of their research. Meanwhile, these are the occasions that could contribute to diversifying the scientific agenda. To do this, it is necessary to discuss the awarding of scientists on the open platforms, in mass media, and develop cooperation with those who are able to use different advocacy formats and can contribute to understanding between those who are in the topic and those outside the "bubble".
4. There is a need to **reposition scientists** and science in the public space, in particular in mass media. This should be an advocacy goal in the interaction between scientists and media and help to build trust of the general public through providing useful, interesting, and important information about science. That is, positioning should be considered as a tool for advocacy, not just popularization.

For international stakeholders

1. To facilitate international advocacy, it is necessary to inform about the opportunities for interaction, **build trust,** and engage scientists in the interaction through motivational programs, which will lead to self-organization in implementation of advocacy projects with the purpose of preserving, developing, and restoring the science.
2. The basic principle of advocacy **"nothing for us without us"** should be applied. Stakeholders should not make decisions on assistance without preliminary communication with Ukrainian experts. Our research has shown that, without this, foreigners have little understanding of the real needs and real situation. For more effective interaction and dialogue, it is necessary to open representative offices of foundations on the ground (Horizon Europe is an example).
3. The key needs are **transparency of criteria**, ease of application and active involvement of scientists in the search for relevant competitions and programs. This will contribute to positive results in providing necessary resources to the scientific community, preserving and restoring its scientific activities in war conditions.

Development of communities requires organizational resources that should be provided for both at the state level and at the level of building an ecosystem of communities in cooperation with other stakeholders.

Introduction

The destructions caused by the enemy lead to new risks, threats, and significant losses for science. In the eyes of the authorities and society, science itself is beginning to look as "secondary" against the background of other more urgent survival needs, although the country's economic stability and its recovery speed depend on its development.

The problem field of science advocacy. During wars and crises scientists have limited access to effective ways of advocating for their needs and impact on decision making on preservation and development of science.

The roots of this problem are:

- limited direct contacts of scientists with government and business representatives, imperfection of existing advocacy mechanisms;
- limited regular exchange of information on the needs of science;
- lack of systematic communication with international partners about the needs of Ukrainian science.

The consequences of hostilities for science and scientists as a vulnerable social group under risk can become threatening without effective and thoughtful advocacy of their interests: ensuring the possibility to exercise basic human rights (safe living conditions, as far as possible under martial law, access to information, for internally displaced scientists — availability of housing and social guarantees, freedom of thought, and personal development), labor rights (to keep their jobs, ensure decent working conditions, etc.) and special rights of scientists (conducting research, access to scientific information, necessary equipment and software, its preservation during the war, possibility of popularizing the results obtained, and professional development).

The internal problems of advocacy are compounded by external postcolonial stereotypes about Ukrainian science abroad which become barriers to defending Ukraine's subjectivity in the world. The problems that existed before the war (lack of systematic communication about the needs of Ukrainian science, low representation of researchers

at conferences, lack of trust in the expertise of Ukrainian scientists, etc.) also hinder the promotion of Ukrainian scientists' interests abroad. Continued Russian influence on some foreign stakeholders is dangerous.

To solve these problems, it is necessary to advocate the activities of scientists at all levels (local, national, international) to the authorities, business, civil society, international governmental and non-governmental organizations and foundations.

In the classical sense, advocacy can have three goals: policy changes, systemic changes, and democratic changes.

Policy changes. In this case, advocacy should take place among representatives of "scientific elites" (scientists with status, research experience, intellectual abilities, and access to resources), government (those who make administrative/political decisions), and business (those who have financial resources for investment and provide the material and technical base). Advocacy can be aimed both at improving the existing mechanisms of preserving, restoring and developing science and at creating new, more effective ones.

Systemic changes require involvement of ordinary citizens (including ordinary scientists who do not hold "high" positions) in the decision-making process. They should help ensure transparency in decision making on the preservation and development of science. At the moment, it is done in the format of a "closed club", mainly through contacts in ministries and such seemingly non-governmental institutions as academies. This requires a reboot of the education system in its relations with science, as well as interaction with the media and other media.

Democratic changes occur when citizens (scientists, students, talented youth, business representatives and other people who want to learn throughout their lives) are aware of their rights to research and demand better access to information about the results of scientific activity. Democratic changes should be covered in media, both in mass media and on the resources of institutions, as well as in thematic blogs of independent influencers. Interaction with the government and business in this context should be aimed at supporting popular science programs in traditional media (TV, radio, press) and popularization platforms, civic science, etc. Implementation of relevant knowledge on innovations is a necessary component of the educational process in the format of organizing hubs, initiating start-ups, etc.

In this study we rely on understanding of **advocacy** as a process of science communication aimed at changing or developing policy, making decisions by stakeholders to improve existing practices to initiate transformations that are beneficial for a particular social group[1].

The **goal** of the study is to identify the needs, existing opportunities and prospects for science advocacy in a period of crisis and to define communication tasks for the preservation, restoration and development of science. To achieve this goal, the following tasks were established:

1. To define the subject field of science advocacy in a period of crisis.
2. To characterize the specifics of communication between scientists and internal stakeholders for the purpose of advocacy.
3. To identify the peculiarities of science advocacy for international stakeholders in a period of crisis (science diplomacy).

[1] V. Usyk, D. Petryk, O. Malchenko (2021) Advocacy for youth councils. Manual. — Kyiv. with the financial support of the British Embassy in Ukraine and the Active Citizens Program of the British Council in Ukraine. — p. 14 *https://www.britishcouncil.org.ua/sites/default/files/advocacy_for_youth_councils_manual.pdf?fbclid=IwAR0w0nbM_OCIh6WTQJ3JajVhJG3GfY6TJG39YW7IzLsik8Aql187FqDgTvU*

Research methodology

The study analyzed the literature on advocacy and communication to support scientific activities in times of crisis.

In foreign studies, the topic of science advocacy was studied extensively and covers the following key topics: the role of society in science advocacy (Mooney, C., 2005; Nisbet MC, Scheufele DA, 2009), public policy and science advocacy (Pielke Jr, R. A., 2007, Cash, D. W., et al., 2003); communication and advocacy of science (Being a Scientist: A Guide to Responsible Conduct in Research, 2009; Bubela, T., et al., 2009); education and science advocacy (Communicating Science Effectively: A Research Agenda, 2017); effective methods of science advocacy (Nisbet, M. C., & Mooney, C., 2007; Sander van der Linden, Leiserowitz Anthony, Maibach Edward, 2019); science advocacy in the digital age (O'Connor, Cailin & Weatherall, James Owen, 2019); social and cultural aspects of science advocacy (Wynne, B., 1996); ethics and science advocacy (Resnik DB, Elliott KC., 2016); science diplomacy (Arnaldi, S., 2016). There are a lot of studies of science advocacy abroad in the context of political research in connection with the actualization of social movements on the problems of environment, biology, etc. (Einstein A. (1947), Tormos-Aponte Fernando et al. (2023), Meyer D. S. (2007) et al.).

In the study we referred to the works on science diplomacy. This is an important area of research related to interaction of science and international stakeholders. A review of the scientific literature on this topic by Krasnyak, O., & Ruffini, P. B. (2020) showed a wide range of studies on communication for mutually beneficial cooperation between the scientific community and external partners. In the works of Börner, K. and Milojević, S. (2019) special attention is paid to the use of the latest methods of scientific data analysis in science diplomacy. Legrand T, Stone D. (2018), Kaltofen, C., Acuto, M. (2018) look into the relationship between science and innovation in the context of foreign policy. A separate report prepared by the Royal Society (2010) addresses current issues in the field of science diplomacy. The work "Science Diplomacy: Foundations and Practice" (2023), which describes the best cases of science diplomats is useful for practical use. The role of science diplomacy in science and innovation in achieving sustainable development is discussed in Flink T. (2021, 2018), Müller, C. N. W. (2020), Mabe, M. & Ashley, E. A. (Eds.) (2017) and Echeverría, L., Aquino, K., & Widmaier, C. (2020). The specifics of science diplomacy strategies and scientific cooperation in specific countries are the subject of study by individual authors, in particular, the national approach of the USA is addressed in Flink, T., & Schreiterer, U. (2010), Spain – Moreno Ana Elorza et al. (2017), China — Li, X. (2023), Latin America — Serafim, M., Bonilla, K., Bámaca-Lopez, E., Castaneda, A., eds. (2021). The partnership between the USA and Cuba is discussed in the work by Ronda-Pupo G. A. (2021). Research cooperation as an important component of French foreign policy can be traced on the official website of the Ministry for Europe and Foreign Affairs. This experience can be very useful for Ukraine, but it is, of course, impossible to mechanically "transfer" it to the Ukrainian context. These practices need to be rethought and adapted to our needs and realities (e.g., "to reboot" the institutions that should facilitate networking of scientists and experience exchange, etc.).

Foreign models of science advocacy prove to be effective, so they can be implemented in Ukraine with account of the existing realities.

The subject field of science advocacy was studied in the frame of expert interviews which highlighted the problematic issues of Ukrainian science under martial law from the point of view of various stakeholders. According to the data of the mass survey, we have a description of the priority needs of scientists in a crisis period, as well as their evaluation of the quality of mechanisms for involving scientists in the decision-making process regarding scientific activities. The analysis of the existing legal framework for science advocacy and the

amendments made to it under martial law, as well as the review of successful cases, allow formulating recommendations on best practices for taking into account the interests of scientists in the decision-making process.

The second task is to determine the specifics of communication between scientists and internal stakeholders for the purpose of advocacy. The results of a mass survey of scientists revealed the best channels for informing about scientific needs, as well as the most frequent ways of informing about scientific requests and needs. Monitoring of the discussion of science in the media allowed identifying popular narratives about effective advocacy of scientific institutions and scientists, and scientific activity in general. The expert interviews provide information about motivation of internal stakeholders and involvement of scientists in solving socially relevant problems.

In the frame of the third task, we studied the peculiarities of science advocacy to international stakeholders in a crisis (science diplomacy). The expert interviews revealed cases of operational coordination of international assistance. This coordination was related to both organization of scientific work and assistance to Ukrainian scientists (those who went abroad and those who stayed in the country). To describe the situation of Ukrainian scientists abroad the corresponding subsample from the mass survey was analyzed.

The fourth task was based on generalization of the collected data with account of different views (triangulation). Thanks to the analysis and synthesis of all the information processes the recommendations were formulated. Thus, during the study the best practices of advocacy in a crisis were described and advocacy opportunities for scientists to promote the preservation, restoration, and sustainable development of science were outlined. The general conclusions will concern the establishment of regular communication with international partners and involvement of scientists in international grant programs.

The following **sources of primary information** were used to complete the tasks:

- **Mass online survey** UAScience.Reload, which included data collection in three waves using a structured questionnaire, allowed comparing the general portrait of the Ukrainian scientist during the Russian-Ukrainian military conflict in three time periods. The third wave of the survey (2,291 questionnaires) specifically included a set of questions about the channels and methods of informing scientists about their needs and solving problems

	1st wave	2nd wave	3rd wave
Online survey activation date	06.04.2022	26.09.2022	04.07.2023
Online survey de-activation date	15.04.2022	21.11.2022	12.08.2023
Number of days to collect data	10	57	39
Total number of questionnaires	2 173	1 729	2 291
Fully completed questionnaires	1 674	1 422	1 879
Partially completed questionnaires	499	307	412

Table. General information about online survey UAScience.Reload [Lutsenko A. et al. (Third Wave Report)]

Detailed results of each of the three waves of the survey are published in three reports [Lutsenko A. et al (Third Wave Report)].

- For **expert interviews**, internal and external stakeholders that belong to the following six categories were identified: scientific community; government institutions; business (including knowledge-intensive business); international stakeholders; media; and civil society. A total of 26 interviews were conducted, including 16 individual interviews and 10 group interviews (several interviewers interviewed the same expert). The preparation of the interview guide was based on an individual approach to each expert, and the logic of questions was built according to their experience and competences which allowed covering a wide variety of cases important in terms of research tasks. The group interview methodology was a useful practice during interviews of experts from government agencies, international organizations, and civil society. Thanks to the selected methodology, it was easier to elicit more detailed answers from the identified categories of stakeholders, as each team member made their own clarifications and thus supplemented the discussion.
- **A semi-structured survey** of scientists (65 questionnaires) complements the mass survey and explains some of the distribution of responses. The survey revealed interesting precedents in the process of interaction with scientists, namely: how Ukrainian scientists requested assistance, who responded and how they helped, what were the obstacles and incentives, etc. An additional survey revealed which practices of interaction with scientists are considered successful, and what kind of assistance scientists needed in the first place.
- *Social media monitoring* of the discussion of science in social networks and online resources during the first year of war (analysis of the array by the Youscan computerized online data collection software). Youscan functionality programmed an automatic text parsing which allows collecting a corpus of messages that correspond to the predefined tag cloud. To form a sample from the existing corpus of 1.2 million messages, 4,000 messages were randomly selected, of which 1,300 relevant messages were selected and analyzed (i.e., those messages that were actually related to scientific topics and scientific work). Based on the data obtained, the discourse of Ukrainian science during the war in online media and social networks was analyzed (February 2022 — June 2023). The lines of division in the discussion of science in society, prejudices, stereotypes, events that influence the public's perception and understanding of science were defined: what do social media users say? in what tone? what values and needs are discussed?

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Chapter 1. Science advocacy in danger

1.1. Contexts and traditions of science advocacy in Ukraine

Wars and crises catalyze all social processes which result in restructuring of systems of interaction between social agents. First of all, this affects communication, which either becomes impossible or, on the contrary, more motivated and reasonable. Unfortunately, this is not obvious either to government officials or business, media and civil society organizations. For the most part, these stakeholders do not understand the value of science, do not take it into account in management, do not implement it in business, and do not cover scientific results in media. The review of the science communication field, with the identification of barriers and opportunities, is the basis for recommendations on science advocacy in Ukraine for its preservation, restoration and development.

As the expert interviews and monitoring data show, public interaction in society (civic activity, politics, relations with the media) is implemented within three strategies of interaction between stakeholders in the scientific process: propaganda, consumer, and activist ones.

The paternalistic strategy was inherited from the USSR as a tradition of determining the value of a particular scientific activity in accordance with the ideological objectives of the ruling party. This strategy is embedded in scientific communication, and even today some institutions keep the forms and principles of hierarchical relationships and following the academic rituals, which creates barriers for including new actors — progressive, competitive, more effective, but who may break the tradition. New generation scientists are critical of these “traditions” [MEDIA01, MEDIA02], and those scientists who had a successful experience of scientific work in the Soviet Union still like the model where the

working conditions were created for scientists and they were not burdened with administrative and financial issues [NAUK02, NAUK09]. In this patriarchal paradigm scientists mostly wait for the right "appointments" and decisions "from above".

The market strategy has flourished thanks to restoration of market relations in Ukraine after independence. It is effective in conducting admission campaigns for higher education institutions and organizing citizen science events, as experts note [NAUK03,04; MEDIA01; BUSINESS01]. The market strategy in Ukraine has helped to make science more understandable to society. However, the understanding of the value of science is changing — not as a space for knowledge production, but as a way of entertainment and profit.

The activist strategy is a strategy of information interaction from person to person which is manifested in grassroots citizen science initiatives, formation of informal networks, platforms, and other forms of cohesion. Unfortunately, without institutionalization and access to decision making to change government policies, such initiatives are marginalized and sometimes even contribute to discrediting certain ideas. Activists in various areas of social relations, including citizen science, promote local knowledge preservation, successfully generate interest in science among the population through popularization methods, and often even fuel consumer interest on the part of business. Meanwhile, these are activist approaches to communication and advocacy that are described as effective by those experts who have a vision and experience of advocating for Ukrainian science to implement changes for restoration, preservation and development of science [NAUK01,02,05; MEDIA04]

Of course, these strategies do not exist in their pure form; hybrid strategies, in which one or the other model prevails, are more realistic. The data from expert interviews show that in communicating with internal stakeholders, the scientific community is more prone to interact according to the propagandistic strategy, that is, in the context of the paternalistic model, to wait for decisions "from above". With few exceptions, scientists are not ready for direct action advocacy. A Ukrainian scientist still has a prejudice against institutionalizing public activity in favor of supporting, restoring and preserving science. But advocacy at the international level requires an institutional approach and knowledge of systemic mechanisms.

As a rule, communication of science is ritualized, which means that it is overloaded with cultural and symbolic functions, which creates barriers to advocating for more effective mechanisms of work of scientists. Of course, there are islands of networking among scientists and science promoters who contribute to its restoration, preservation and development. However, this activity has the character of individual cases, and there is a risk that it will not turn into sustainable development programs after the war.

1.2. Regulatory framework of science advocacy in Ukraine

The Constitution of Ukraine contains the key provisions that establish the principles of advocacy in Ukraine. In addition, the issues related to advocacy are regulated by a number of **laws of Ukraine,** including the Laws "On Citizens' Appeals", "On Access to Public Information", "On Prevention of Corruption", "On Principles of State Regulatory Policy in

Economic Activity", "On Public Associations", and others.

For a long time, there was no separate legal act that would regulate advocacy (both in general and science advocacy) in Ukraine, despite the fact that the Anti-Corruption Strategy for 2014-2017 contained a direct commitment to adopt legislation to define the legal basis for lobbying. The relevant draft laws have repeatedly been submitted to the Parliament of Ukraine (since the late 1990s), including two draft laws on lobbying in 2016 and a draft law on public advocacy in 2017, but none of them has been adopted.

Adoption of legislation on lobbying was one of the EU's requirements for the start of negotiations on Ukraine's membership in this organization. On December 7, 2023 the government approved the **draft law "On Fair Lobbying and Advocacy in Ukraine"** and submitted it to the Ukrainian Parliament for consideration. According to the drafters of the law, "the document distinguishes between lobbying and advocacy by three criteria: profitability or non-profitability, advocacy of private or public interest, and identification of funding sources. Lobbying is carried out on the request of private companies or individuals for profit. The purpose of advocacy is not to make a profit. The source of funding may be, for example, international partners of Ukraine. Advocacy is carried out by non-governmental organizations"[2].

The Law "On Scientific and Scientific and Technical Activities" (2015) defines the rights of scientists (Article 5) and researchers (Article 6). Article 38 defines additional rights of researchers to social protection. Unfortunately, exercising these rights is not effective and requires advocacy. There are often situations when the rights of scientists are interpreted as obligations in universities, institutes and other scientific institutions. For example, academic staff are required to fill in plans and reports with information on participation in conferences, publications in scientific journals, etc. The Law "On Scientific and Scientific and Technical Activity" regulates the status of public scientific organizations, one of their tasks is to protect professional interests (Article 23), and councils of young scientists, which are formed to ensure protection of the rights and interests of young scientists (Article 24).

Changes after the beginning of russia's full-scale invasion

After the beginning of russia's full-scale aggression against Ukraine in 2022, the legal framework of advocacy in general and science advocacy in particular has undergone certain changes. According to the Decree of the President of Ukraine "On the Introduction of Martial Law in Ukraine" (24.02.2022), the provisions of the Law **"On the Legal Regime of Martial Law"** came into force in Ukraine. In particular, the martial law regime provides for:

- restrictions on movement within the state and travelling abroad,
- introduction of certain peculiarities of regulating labor relations,
- possibility of seizing property in private, communal and state ownership,
- regulation of information exchange,
- possibility of restricting the provisions of Articles 38 and 39 of the Constitution related to citizens' participation in management of public affairs, as well as their right to peaceful assembly and demonstration.

The mentioned norms do not only set restrictions for advocacy of science, but also affect the process of scientific and scientific and technical activities in general. The question arises whether it is possible to create special legislative conditions for scientists and scientific institutions during war.

[2] URL: *https://forbes.ua/money/ukraina-gotuetsya-do-legalizatsii-rinok-lobistiv-blitsintervyu-z-golovoyu-nazk-novikovim-pro-noviy-zakonoproekt-ta-klyuchovi-zmini-dlya-biznesu-04122023-17653*.

It is for this reason that the Law of Ukraine **"On Amending Certain Laws of Ukraine on State Guarantees During Martial Law, Emergency or State of Emergency"** (15.03.2022) introduced additional state guarantees for researchers and academic staff who had to change their place of residence under martial law. They were guaranteed to keep their jobs and average wage, and, if necessary, to be provided with accommodation and meals.

A number of laws adopted during 2022 clarified the rules on the **organization of labor relations** under martial law ("On the Organization of Labor Relations under Martial Law" as of 15.03.2022, "On Amendments to Certain Legislative Acts of Ukraine on Optimization of Labor Relations" as of 01.07.2022).

During the interviews, experts repeatedly mention that in a number of cases, **the regulatory framework for scientific and technical activities** has been simplified. It should be noted that the introduction of martial law in Ukraine has activated Article 55-2 of the Regulation of the Cabinet of Ministers of Ukraine (CMU), according to which the Prime Minister or another member of the CMU, with the consent of the Prime Minister, may submit a draft act to the CMU to address urgent issues without complying with the requirements for all necessary approvals and consultations. If approved, this act is finalized by the CMU Secretariat without changing its substance[3]. This article allows to speed up consideration of issues related to protection of scientists' lives, their safety, and improvement of conditions during martial law[4].

1.3. Emergent communication during the war

The respondents noted that the most effective method of advocacy was **interpersonal communication** — meetings with city residents, communication with colleagues, journalists, representatives of ministries and agencies, the territorial defense and the Armed Forces of Ukraine.

Email and messengers became the most optimal means of communication within the scientific team to coordinate actions during the military action [NAUK01]. The Russian military could check the phone at any time and see subscriptions and messages that indicated the scientist's pro-Ukrainian position, so they deleted chats and emails, used old push-button phones, and tried to choose the most secure messengers [NAUK01, 05, 08]. Teachers and students of higher education institutions **used local chats** of departments and student groups, and there was no single channel from the university. The example of Pryazovskyi State Technical University from Mariupol is illustrative, it did not have a

[3] URL: *https://zakon.rada.gov.ua/laws/show/950-2007-%D0%BF#n3705*.
4 Expert interview of Igor Taranov.

complete communication chain before the full-scale war, that's why issues of interaction and coordination had to be resolved in the emergency mode [NAUK07]. The experience of the occupation of Mariupol shows the need to create a complete dossier with the addresses of employees and backup options for contacting them. We cannot allow the documents, seals and digital signatures to fall into the occupiers' hands.

In some cases, it was possible to convene an emergency **meeting of the university administration** [NAUK06]. For example, at Sumy National Agrarian University, members of the university administration gathered in the rector's office at 7 am on February 24. Direct communication was more effective, fast and accurate [NAUK04]. Direct orders helped to communicate operational decisions smoothly and avoid chaos. Thanks to a well-coordinated group chat system, it was possible to spread the recommendation to teachers and students to stay at home within an hour [NAUK04].

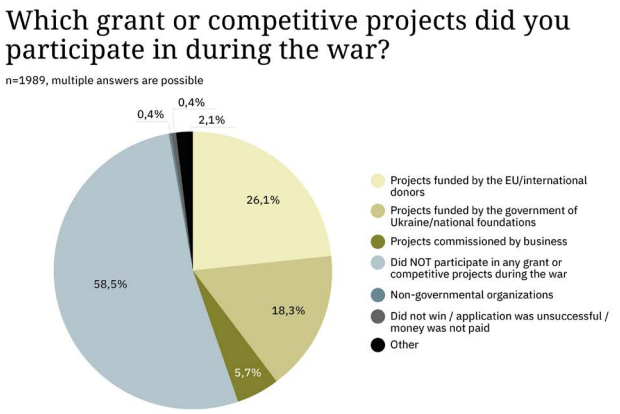
The existing communication chain within the university helps convey the key message to everyone in a crisis. The communication scheme is based on the following hierarchy: rector — vice-rectors — deans — a) heads of departments — teachers, supervisors, laboratory assistants; b) supervisors, dormitory managers — students. The open position of the rector was expressed in an address to the staff, recorded in a video and published on the university website [NAUK04]. This case shows to be effective, but the question is whether it can be an example. It is a very complex hierarchical scheme, and general chats would be faster, but the administration is often afraid of "uncomfortable" questions from students and teachers. Thus, the system is designed to "bring" information about needs through a bunch of intermediaries. In the cases where there was no single chain, it was much more difficult to ensure communication interaction of the scientific team.

The war notes of scientists (and sometimes universities) in social media helped disseminate information about the needs of scientists and students. Readers helped to solve many problems (for example, people who stayed under occupation were helped to resume salary payments). Today, these subscribers also read posts about science, research, etc. However, in the first weeks of the great war and during the occupation, a lot of scientists temporarily closed their pages. It was dangerous to publish pro-Ukrainian posts or simply express their views.

Ways to communicate scientific requests and needs

In addition to channels for communicating their scientific needs, it is worth identifying effective means of communicating a key request for a research project. Scientists prefer personal communication with those who can provide necessary resources and protect their rights (47%). In second place, scientists ranked participation in grant projects from international foundations and organizations (40.2%), which generally provides opportunities to receive direct funding and an ability to manage their own budget.

Participation in conferences and other scientific events in Ukraine (25.6%) and applying for Ukrainian grant projects (19.5%) are becoming an important way to express scientific needs. Those who have never received targeted assistance from the state or won grants for research projects are critical of these forms of meeting the research needs, in particular, the survey expresses the opinion that "Only those who have received grants before receive them. I have never received any assistance from the state for research, nor from organizations." Next are appeals through the public, namely, publicity in social networks/messengers (19.1%) and mass media (12%). The unpopularity of this method of informing is related to fears of being fired. Only 18% of respondents rely on the form of an official appeal [Lutsenko A. et al (Third Wave Report)].



Distribution of answers to the question "Which grant or competitive projects did you participate in during the war?" (n=1984) [Lutsenko A. et al. (Third Wave Report)]

During the war, Ukrainians often switch their activity from research projects to volunteer, media projects, civic initiatives, and military purposes. In a crisis situation, scientists continue to prepare applications, submit documents, and participate in competitions for funding for research projects, educational projects continue, and there is exchange between partner universities. It is quite reasonable that some applications do not meet the formal requirements and research projects do not win in competitions. However, as experts mentioned in their interviews, it is not normal when the winning projects do not receive the promised money, or even if they do, it is too late [NAUK02, 04]. One respondent noted: "We won international projects, but the money has not come, and we are working on them because they promise money (because it is under the UN)." Ukrainian scientists continue to work in the hope that financial transactions will eventually take place. It should be noted there are cases of self-funded initiative projects.

An important observation is that national programs to support research projects, including budgetary funding from the National Academy of Sciences of Ukraine, the National Academy of Medical Sciences of Ukraine, state budget research, grants from the National Research Foundation and others, as well as private charitable foundations (Academy of Ukrainian Press, International Academy of Environmental Law, foundations, representative offices of the Greek Catholic Church, etc.), as well as government orders become significant support for Ukrainian scientists. However, interaction between institutions and individual scientists is effective where there is a good information and communication exchange. Where communication is disrupted, advocacy activities to support science risk being aimed not at solving the problem, but at eliminating the institutions themselves, which leads to a decrease in trust in science, regardless of whether individual departments or scientists are effective or not.

1.4. Needs communicated as priorities in a crisis

One of the key principles of advocacy is participation of those who are directly involved in the problem. The cycle of the advocacy process starts with clarifying the problem, finding solutions, identifying stakeholders, and then moving on to formulating messages [Advocacy for Youth Councils (2021); Politically Smart Advocacy (2018)]. Experience shows that advocacy is effective when the key message is formulated in conjunction with the need.

In the expert interviews, scientists motivated their actions to meet the needs as follows:

- keeping the team;
- saving the research;
- keeping the employees' jobs;

- saving the laboratory, equipment;
- publication of research;
- recognition of a scientist abroad;
- saving lives (comes to the fore in a crisis).

An online survey allowed **detailing the needs** that motivate scientists to act, as well as finding out what blocks their initiative [Lutsenko A. et al. (Third Wave Report)]. Thus, the top five needs also include financial, safety, psychological, social and household needs, which people cannot ensure themselves.

Receiving salary

While salaries are mostly paid (86.7%), there are definitely problems with the regularity and scope of these payments, and with the reduction of salaries. Scientists sometimes have to agree to small parts of their salary (12.5% of the researcher's salary; 0.7 of the lecturer/associate professor's salary, etc.) [Lutsenko A. et al (Third Wave Report)]. The labor rights of scientists are protected on the whole, but violations are not excluded, and scientists consider it pointless to fight them and do not always start. It is not uncommon to not receive holiday pay and financial assistance for health improvement. In some institutions, scientists are forced to take leave at their own expense, and there are unfair dismissals.

As the examples show, people who work by vocation are motivated. Unfortunately, without proper working conditions and remuneration for work, there is a risk of burnout or marginalization of the scientific community.

Working conditions and opportunities for scientific activity

There are positive trends in the ability of scientists to be present at the workplace where they can conduct research. In spring 2022, the premises of scientific institutions were heavily damaged as a result of shelling, and every third scientist was unable to visit their laboratories and offices. In the second half year of the war (autumn 2022), a fifth of the surveyed scientists were not able to continue their research at their workplace. In the summer months of 2023, only 17.1% of respondents mentioned this problem. A lot of scientists' homes where their personal research archives were stored were destroyed. For example, one respondent noted that he *"lost computer equipment, his apartment burned down, part of the material collected for research was lost, and a third of his personal library burned down"*. [Lutsenko A. et al (Third Wave Report)].

The respondents' mentions of overcoming various bureaucratic obstacles deserve special attention. Among the reasons that prevent them from doing science, scientists pointed to the restrictions of the administration: lack of interest in the scientific work of employees on the part of the management (*"harassment by the management, public humiliation for going abroad")*, workload with routine meaningless tasks *("a heavy workload with nonsense at work which requires considerable effort, but have no prospects")* [Lutsenko A. et al (Third Wave Report)].

The respondents repeatedly mentioned bureaucratic barriers: *"Overload with paper and electronic reporting, excessive control, simultaneous provision of full-time and online education, endless accreditations that exhaust the teacher and a huge amount of unpaid work done"* [Lutsenko A. et al (Third Wave Report)]. Systemic reasons related to martial law were also mentioned. One respondent raised the issue of paying 20% VAT when receiving financial support for an international project. *"No one from either the National Academy of Sciences or the Ministry of Education and Science could adequately explain the steps, the algorithm for properly registering a project in the legislative field of Ukraine in order to avoid paying such a tax from an international grant because its payment should not be covered by grant funds"* [Lutsenko A. et al (Third Wave Report)]. The respondents also emphasize

the problem of the high cost and specificity of peer review of articles for publication in internationally reputable journals. Financial and bureaucratic problems cause the thoughts of the lack of prospects for scientific work for young scientists. *"The lack of opportunity to work on modern equipment, constant cuts in research and development rates ... makes us think about the lack of prospects for further work in this area and is the reason for the outflow of young staff with whom you can really do something useful and relevant"* [Lutsenko A. et al (Third Wave Report)].

As for the relations with the authorities, experts repeatedly express the opinion that the public administration system does not help but creates additional barriers to preservation, restoration, and development of science. According to one of the experts, Acting Director of O. Ya. Usikov Institute of Radio Physics and Electronics Yuri Logvinov, *"the management system is designed in such a way that it cannot do anything else except for imposing various kinds of punishment. It is necessary to change the principles of the authorities' work, and it will probably be more difficult to do than to defeat the Russian Federation".* [NAUK02] Some facts about violations of the rights of scientists or obstruction of scientific activity do not appear in the public space, but are spread in personal discussions. For example, there are cases when in October and November 2022 the tax service imposed fines for overdue payment of taxes and fees, and the prosecutor's office selectively began checking the state of the research institution's shelter, etc. (see Annex "*Case folder*").

In wartime circumstances, bureaucratic reasons only complicated problems: in addition to the objective conditions of uncertainty, destruction and danger, one has to take into account orders from ministries and departments which are not actually implemented because they become irrelevant in a crisis or cannot be fulfilled. This leaves scientific managers face to face with crisis management [NAUK06].

An illustrative example of the government's incompetent policy is keeping formal requirements for funding scientific institutions. Thus, the Ministry of Economy's assistance program for research institutions has proved to be a non-functional mechanism for returning money in the form of profit within an investment project. In general, the proposed form corresponds to the realities of scientific work as research institutes are profitable organizations. However, it requires preparation of an expensive (UAH 2–3 million) independent assessment of the damage, development of design and estimate documentations, and a procurement plan, which is impossible in the context of reduced budget funding during the war. *"If we had this money, we would not ask for help, but would do everything ourselves"* [NAUK02]. In spite of the fact that scientists tend to keep silent about their current problems, considering them unimportant compared to other challenges in society, the experience of living through the crisis makes scientists emphasize the need to articulate specific requests and record actions to transfer knowledge on how to overcome the crisis.

Evacuation of scientists and scientific equipment

During the war, the situation changed rapidly and unpredictably. The expert interviews showed that the algorithm of actions proposed by some agencies (for example, the Ministry of Culture and Information Policy) did not work, it collapsed, in fact. Employees were unable to contact state departments, had no access to the Internet, no electricity, no way to get to institutions, no transport (and often no instructions) to evacuate people, books, and museum treasures. Individual means of communication played a decisive role, which required personal expenses and resources.

In the first days of the war, scientists were in dire need of evacuation organization [NAUK08]. The people who stayed on the occupied territories were in an information vacuum, with no means of communication to coordinate their actions.

As one of our experts says: *"During a war, you can only rely on yourself, your own intuition and instinct for self-preservation. There is no protocol, no "recipe" for rescue, no one to expect help from"* [NAUK07].

Relocation of an HEI is formalized by an order of the founder (Ministry of Education and Science, other ministry, owner of a private institution) and is displayed in USEDE (Unified State Electronic Database on Education) as a change of actual address. The procedure for relocating a university is as follows: hold an online meeting to assess the situation; negotiate with the head of the host institution; make a decision to approve the new address; contact the Ministry of Education and Science; describe the actual situation; re-register the actual address of the HEI.

The experience of Pryazovskyi State Technical University shows several points that are recommended to be taken into account for crisis management: a format of remote work of the administrative part (accounting, legal department, planning and financial department), close communication with an executive authority (organize meetings and consult with the tax service, treasury), the Department of Social Welfare); providing access to information resources and platforms (library, laboratory, methodological support) [NAUK07].

Later there was the task to restore the destroyed buildings and laboratories. Scientists looked for resources on their own: they asked the military for demining and the management of the research institution to repair the material and technical base [NAUK05]. Most of them relied on their own strengths and capabilities and saw no point in asking for help from other stakeholders.

Gender specifics that should be taken into account in determining the needs of scientists

At the beginning of hostilities, scientists, regardless of their ranks and qualifications, had to perform tasks that often did not correspond to their job responsibilities. In addition, in the expert interviews, as well as in messages collected within social media monitoring, it was repeatedly noted that during the war, male scientists may be completely excluded from scientific activities due to their presence at the frontline or have limited access to international cooperation due to a ban on travelling abroad. There are frequent reports in the media and social media that young scientists are fighting and dying at the frontline. The needs of scientists who are elderly or need inclusive workplace conditions and labor standards are also mentioned.

The findings of sociological surveys show that female scientists, on the one hand, are overloaded with household problems as they are traditionally responsible for taking care of children, elderly relatives, etc. During the war, the number of displaced female scientists is higher than that of men, and this is because of the decision to relocate to a safer place [Lutsenko A. et all (2023)]. On the other hand, a window of opportunity has opened for women to represent Ukrainian science abroad. As a rule, if women go abroad for scientific activities, they receive better working conditions compared to the ones they had in Ukraine. Experts in the social sciences have repeatedly emphasized that women's return to Ukraine is possible if not only basic needs but also educational and infrastructural needs for them and their families, for which they have taken responsibility, are met.

To sum up this chapter, it should be noted that during the war or in a crisis, basic needs come to the fore, and society ceases to see the strategic perspective behind tactical actions, which includes science, education and culture. Desk research and analysis of expert interviews within this project have shown that:

- during the crisis, scientists act under uncertainty, often take responsibility and make decisions faster than representatives

of the relevant executive departments on the ground, act not only to preserve science but also to save lives; storytelling about this can be the basis for positioning science and scientists for public support;

- advocacy during the crisis was mainly related to tactical rather than strategic issues of preserving science in Ukraine;
- advocacy is based mainly on personal communication of the scientist, and the most effective in communication were the institutions that already had teamwork in place, a network, and were familiar with the methods of working in a crisis;
- there is an understanding that the needs of scientists differ due to age, status, and gender differences, and meeting these needs requires specific advocacy mechanisms.

Unfortunately, scientists have a lack of understanding of advocacy effectiveness, and the mechanisms for protecting and promoting the interests of the scientific community during the war are even more limited. Mechanisms for advocating for the needs of scientists do not work systematically, and the result usually depends on personal acquaintance or chance. This does not give confidence in the sustainable development of science both during the crisis and after overcoming it.

Chapter 2. Science advocacy to internal stakeholders in danger

2.1. Scientists and teams of scientists as subjects of advocacy

The analysis of expert interviews allows outlining the communication patterns typical for scientists. A scientist communicates personally. He or she is recognized by partners and colleagues. The components of a scientist's personal influence are his or her research, connections, and publications. The scientist takes responsibility and makes decisions about the future.

The experience of scientists' activities during crisis, their statements about the motivation of these or those decisions and actions during a military threat, allows outlining the image of a scientist. Scientists have a vision that allows them to be described as a community of like-minded people. Besides, society has expectations about scientists which are outlined in the documents on the competences of a researcher.

The focus group discussions with the experts of this project outlined the ideal of a scientist. A scientist is a person who:

- produces knowledge. Providing access to this knowledge is necessary for the development of society;
- values academic freedom;
- creates an environment around him/herself;
- tends to be self-organized;
- takes responsibility;
- has a strong communication potential;
- has unique experience, competences, skills that allow him/her to make decisions independently and act in conditions of uncertainty. To have these competences, a scientist acquires knowledge over a long period of time — it is a slow investment that ensures the production of knowledge.

The mentioned qualities can become a basis for **positioning of scientists** in advocacy and popularization of science. It is not always possible to meet this ideal in practice, but having it as a vision is useful for developing the positioning of the scientist's image during information, popularization, and advocacy campaigns. During crisis, these qualities are the ones that are the most noticeable, allowing making decisions and changing the world around us. Communication with scientists within expert interviews makes it possible to say that scientists are confident that science is a social value, but they do not always articulate this confidence in public communication with those who do not share their values.

As a person who has knowledge and is not afraid to take responsibility, a scientist can assess the consequences and predict the future, create an environment and offer solutions to problems, and act effectively and confidently. At the same time, these very skills lead to rebooting the scientist during a crisis. Scientists are also expected to be able to administer projects, and the administrative burden eats up the time needed for scientific work — knowledge production.

As the examples from the expert interviews show, **hierarchical relationships in teams** sometimes allow for quick decision making, but there can often be delays in the implementation stage, as each step requires approval, which overloads the leader. In horizontal teams, decisions take longer to make, but the implementation stage produces more sustainable results.

The scientist's active civic position helped find ways to effective methods of advocacy. *The first step* is to articulate problems and share them through social media, *the second* is to gradually get to know and join the communities that solve similar problems. Thus, the teams set themselves similar goals and followed the same strategy. *The third step* is related to increasing the interest of policy makers, when there is an opportunity to express their ideas and explain the necessity (or irrationality) of management decisions. Such initiatives as Science Days and Science Picnics received positive feedback. They have become platforms for an equal and effective dialogue between scientists and the public. The fourth stage is participation in lawmaking, discussion, lobbying, and promotion of ideas for reforming scientific activities [CIVIL01].

Observations show that when a scientist advocates for the needs of academic activity on their own, it has risks — burnout and inability to pass on the experience to others. In addition, treating your own experience as a unique individual case becomes a barrier to outscaling of best practices by scientists. Thus, individual efforts will have less effect than collective efforts, including networking and coalition building. During the crisis, **network advocacy** becomes more relevant. Stakeholders are under a strict time limit and listen either to the most authoritative individuals or to powerful coalitions.

The study shows that in the context of preserving, restoring and developing science during a crisis, in the context of advocacy the following can become the basis for building communities:

- **scientists and their teams** (departments, working groups, including postgraduate or student groups);
- **scientific centers** of science popularizers and their supporters (scientists-educators and the circle of communication — students, like-minded people, parents);
- **networks of scientists** with scientific interests (international, national, local).

The analysis of interviews with scientists and science mediators showed that a barrier to the implementation of advocacy activities is the unwillingness and inability of advocacy actors to identify or name the need that must be met to preserve, restore or develop science. This is a mental barrier that is caused by both the simplicity of the need (ensuring decent working conditions, such as availability of toilet, heating, lighting, free access to a laboratory or educational building, etc.) and the reluctance to initiate changes that may have administrative consequences (inefficient work of the HR department, legal department, accounting department, etc.). The formulation of advocacy goals is also difficult, as it is not always (mostly) within the scope of a scientist's research interests.

2.2. Key stakeholders in science

For effective advocacy, it is necessary to clearly identify “targets” in the government. That is, those entities that can influence decision making to meet a particular need of society. They are called stakeholders. In accordance with the tasks set, a **general map of stakeholders** in scientific and technological activities was formulated. It should be noted that responsibility of each agency does not cover the field of science as a whole.

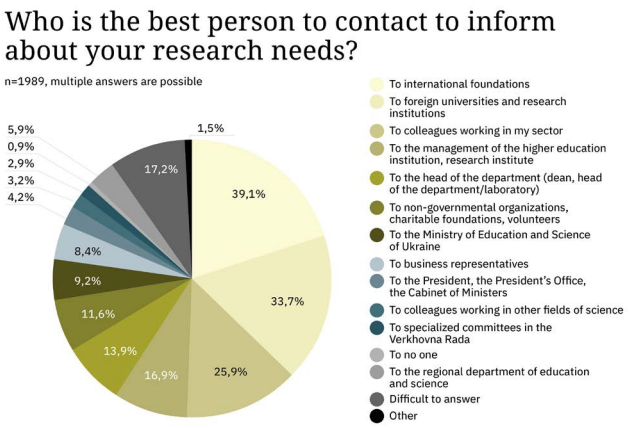
Thus, the stakeholders of scientific and technical activities include:

- President (President’s Office)
- Cabinet of Ministers of Ukraine
- Ministry of Education and Science of Ukraine
- Verkhovna Rada of Ukraine (Ukrainian Parliament), in particular the Verkhovna Rada Committee on Education, Science and Innovation
- Local governments, their departments of education and science
- Higher education institutions
- National Academy of Sciences and national sectoral academies of sciences
- National Research Foundation of Ukraine
- Scientific institutions
- Business
- Charitable organizations
- Crowdfunding initiatives
- International foundations
- Foreign universities and research institutions
- Trade unions, including the Trade Union of Education and Science of Ukraine
- Non-governmental organizations
- Volunteers who help on an individual basis
- Friends, relatives, colleagues (given that individualized communication in disseminating information, sharing experiences and advocacy through social media is currently playing a significant role).

Based on this list, scientific research institutions should adjust their stakeholder map depending on specific problems and interests. This map (preferably with the contacts indicated) should always be available at the scientific institution so that in case of need or even emergency, all possible options whom to seek support from are visible.

The survey of scientists showed that **scientists’ trust in different stakeholders** and willingness to engage with them are very different.

Below is the distribution of answers to the question “Who is the best person to contact to inform about your research needs?” (n=1989, multiple answers are possible) [Lutsenko A. et al (Third Wave Report)]



The scientists would most often choose international foundations as recipients of information about scientific needs and protection of their rights (39.1%). One third of respondents (33.7%) would inform foreign universities and research institutions and seek their assistance. One in four scientists (25.9%) would discuss their problems and needs with colleagues working in the same field of science [Lutsenko A. et al (Third Wave Report)].

According to the respondents, the academic institution where the scientist works directly, primarily the institution's management, administrative department (e.g., the international department of the HEI) (16.9%), and the head of the department (dean, head of the department) (13.9%) are an important stakeholder [Lutsenko A. et al (Third Wave Report)].

The quite reasonable expectations about the primary interest of the state in scientific developments of local scientists ("First of all, the state should be interested in its scientific potential, and then all the others, including business") do not always correspond to the real situation. The online survey data confirm the low potential of addressing the authorities as an advocacy tool for scientists. The authorities that deal with scientists in one way or another are addressed in the second turn (the Ministry of Education and Science — 9.2%, sectoral committees in the Verkhovna Rada — 2.9%, local authorities — 0.9%). Only 4.2% of respondents rely on the guarantor of the Constitution and higher executive authorities. Some 5.9% of respondents do not expect any special help from anyone. [Lutsenko A. et al (Third Wave Report)].

An active part of scientists take the initiative and rely only on themselves, finding funds, participating in competitions, receiving grants or simply using their own funds, asking friends and colleagues for help.

Some respondents do not believe not only in the effectiveness of appeals related to solving the issues of science management, but also in the possibility to defend the rights of scientists in a democratic way. There is a skeptical attitude to the possibility of getting any help. There are opinions that it is inappropriate to talk about science during the war, as the priority is to solve economic and legal problems in the country.

As we can see, there is a need to create mechanisms that would help to connect the capabilities of the authorities and make optimal use of them for effective governance. Such mechanisms may include wide coverage of successful practices of interaction between government agencies and scientific organizations, regular rotation of staff in committees and commissions responsible for the development of science, informal feedback from government representatives and transparency of management decision-making procedures, expansion of the list of open data mandatory for publication and control over their implementation, etc.

2.3. Advocacy of science to the authorities during martial law

Key actors in science advocacy

Scientists in political and civil service positions

One of the most effective forms of science advocacy can be the entry of scientists into political and civil service positions (hereinafter — public positions) with further development and implementation of necessary measures to meet the needs of the scientific community. As noted by the respondent scientists working in government and public institutions, their motivation to

take this job was the desire to change the regulatory framework related to funding of research projects, create a friendly ecosystem for scientific and scientific and technical activities, and stimulate the development of science by eliminating existing limitations and difficulties[5].

In this context, **intra-agency and inter-institutional advocacy** is important when a science advocate working in government has to promote and defend the needs of science to other departments or government institutions.

The speed of decision making and result of their implementation in practice depend on the motivation of a specific person who has influence due to his or her position or status. This is especially important for the preservation and restoration of science in the context of the full-scale Russian invasion. The situation with the Ukrainian-Swiss Joint Research Projects competition was illustrative: although the Swiss National Science Foundation was ready to allocate CHF 5 million for the competition in early summer 2022, the necessary CMU resolution that would enable the NRFU to carry out the relevant activities was adopted only a year later, on 4 July 2023, i.e., after the new team had arrived at the MES.

Experience shows that **scientists who want positive changes** in the development of Ukraine's scientific sphere are not sufficiently involved in the implementation of their proposals in the field of public administration.

National Academy of Sciences of Ukraine

According to the NAS Statute (Article 11), one of its main tasks is to protect the rights of NAS members and employees of its institutions[6]. Although this provision does not refer to advocacy of science in general, however given the large number of scientific institutions subordinated to the NAS and their researchers (over 26,000 people), the advocacy activities of the Academy cover a significant segment of Ukrainian science.

Thanks to its special status as the highest self-governing scientific organization in Ukraine and a non-profit state budgetary institution[7], the NAS has the opportunity to participate in the process of public policy making. However, in an expert interview, Anatoly Zagorodny, President of NAS, noted that changes to the legislation regulating the field of science in general and the activities of the NAS in particular are very often adopted without discussion and consultation with the Academy, without involving its experts[8].

The NAS advocacy activities during the martial law were focused on three key areas. First of all, it was aimed at solving the problems of those scientific institutions that appeared in the combat zone and suffered significant damage. To help the institutions in the combat zone, the Academy cooperated with the heads of regional military administrations, in particular in Kharkiv and Odesa. The NAS also raised the issue of practical implementation of defense developments created by representatives of scientific institutions with the Ministry of Defense of Ukraine and the General Staff of the Armed Forces of Ukraine. Finally, another area is countering the ideas of transforming the NAS into a non-governmental organization and taking away its management functions[9], which are contained in the document "Vision of Ukraine: Reform and Restoration of the Social and Humanitarian Sphere in the Perspective of 2030". The NAS representatives held a series of discussions of these proposals, in particular in the pages of the newspaper "Svit"[10] and in

[5] Expert interviews of D. Kurbatov and I. Taranov
[6] URL: *https://zakon.rada.gov.ua/laws/show/n0001550-02#Text*.
[7] The Law on Scientific and Scientific and Technical Activities, Article 17
[8] Expert interview of A. Zagorodny.
[9] Expert interview of A. Zagorodny.
[10] URL: *https://svit.kpi.ua/2023/08/01/анатолій-загородній-оновлення-систе-2/*

person with participation of people's deputies of Ukraine[11]. During the discussions, the proposals set out in the "Vision of Ukraine" had much criticism from the NAS. This situation is an example of how different interests of stakeholders lead to opposing proposals within their advocacy activities.

Observations show that **active and systematic cooperation with the civic sector** on a partnership basis, including volunteer initiatives, charitable foundations, and NGOs, allows for joint decisions, joint implementation of projects to meet the needs of science and advocate for necessary legislative changes.

National Council of Ukraine for Science and Technology Development

The National Council of Ukraine for Science and Technology Development, established under the Law of Ukraine "On Scientific and Scientific and Technical Activities", has remained the highest advisory body to the Cabinet of Ministers of Ukraine during the martial law period, dealing with the formation and implementation of the state policy in the field of science. The National Council comprises the representatives of higher education and scientific institutions, as well as representatives of state authorities led by the Prime Minister involved in scientific and technical activities. All these make it an important platform for advocating for the needs of science at the highest level.

However, the problems that characterized its activities in the previous period became even more pronounced after February 24, 2022.

First, the National Council meets irregularly — in four years, only two meetings were held in 2020–2023, including from the moment the martial law was introduced (February 2022) to the present time (November 2023) — the only meeting was held in February 2023. The absence of regular meetings means that members of the National Council have to rely mainly on personal connections with ministries and agencies to address the needs of science[12].

Second, the Administrative Committee of the National Council has not begun its full-fledged activities yet, so the key functions of the National Council are performed by the Scientific Committee, which unreasonably increases the workload of its members.

As for the **Scientific Committee** of the National Council, its advocacy activities are noticeable and, in some cases, cause a wide public outcry. In particular, the *proposals* of the Scientific Committee to reform the sphere of scientific research, development and innovation raised active discussions[13]. In addition, the Scientific Committee was involved in advocating for amendments to regulations to launch the Ukrainian-Swiss Collaborative Research Projects competition.

However, there are certain drawbacks in its activities. Some of the scientists who are elected to the Scientific Committee are not ready to work with the authorities and do not understand how the "bureaucratic machine" works. The committee members spend the first one and a half to two years of their term in office only to understand the specifics of lawmaking work[14].

The level of involvement of the scientific community in interacting with the Scientific Committee is also low. According to Oleksiy Kolezhuk, the Chairman of the Scientific

[11] URL: *https://www.facebook.com/permalink.php?story_fbid=pfbid02N97vZ5kAtUCRo1BRCZ5vxEajEnUcrxj4scYGhhbZENairJ8WoZpem6LgFHrFBUW4l&id=100038414404277*.
[12] Expert interview of O. Kolezhuk.
[13] URL: *https://sci-com.org.ua/wp-content/uploads/2023/07/%D0%9F%D1%80%D0%BE%D0%BF%D0%BE%D0%B7%D0%B8%D1%86%D1%96%D1%97-%D0%A0%D0%B5%D1%84%D0%BE%D1%80%D0%BC%D0%B0_%D0%BD%D0%B0%D1%83%D0%BA%D0%B8_%D0%9D%D0%9A_%D1%87%D0%B5%D1%80%D0%B2%D0%B5%D0%BD%D1%8C_2023_final.pdf*.
[14] Expert interview of O. Kolezhuk.

Committee (until October 2023), *"in 6 years, there have been literally a few appeals when people "raised" a real, general problem."* Moreover, since the beginning of the full-scale invasion, the Scientific Committee has not received any requests from scientists about their needs[15]. This shows that the scientific community is not sufficiently aware of the advocacy capabilities of the Scientific Committee.

Another problem is **interagency disagreements**, when some government officials perceive the National Council and, in particular, its Scientific Committee as competitors who take away some of their functions and powers. *Oleksiy Kolezhuk* notes that they "systematically worked against us, which, unfortunately, has borne some fruit in terms of the attitude of some high-ranking officials to us".

Trade union of science and education workers

The trade union of science and education workers, both national and local it branches which operate in HEIS, is an important stakeholder in science advocacy. Protecting the rights of students and researchers is the statutory goal of its activities. Unfortunately, the experts in the interviews never mentioned the experience of turning to trade unions in crisis. This does not mean that there was no work to be done. Trade unions mainly work as a mutual aid, and the possibility of getting assistance depends on the personal relationship of the scientist with the leadership of the branch. An analysis of the website *https://pon.org.ua/* shows that information about the organization's activities is mostly formal, and we have not been able to find any information that would show that the trade union protects the labor rights of scientists.

Public organizations and initiatives

The academic community is ready to engage in discussions, collaborate, and take initiative to protect its rights and express its needs. In June 2023, the School of Political Analysis of the National University of Kyiv-Mohyla Academy and the Reanimation Package of Reforms Coalition prepared a vision document "Vision of Ukraine: Reforming and Renewal of Social and Humanitarian Sphere by 2030"[16]. It states, among other things, that the scientific community has a weak self-organization and is not able to counteract, for example, the imitation of science (based on false incentives for a scientific career, fulfilment of formal requirements for obtaining titles and degrees, etc.).

However, *Anton Senenko* disagrees with this statement and believes that scientists are quite self-organized, and the problem lies in the biased leadership. "All this reformist "scientific junta" began to self-organize under Tabachnyk, when we opposed Order No. 1112 "on 5 articles" (i.e., the need to have five articles for thesis defense) and other nonsense. This party from Science Days and Science Picnics still exists. There are meetings called "Weekdays of Science", and these people formulate certain proposals"[17].

Science mediators often express a skeptical view of the effectiveness of advocacy. Anton Senenko says that *"regarding reforming the scientific sphere: university reforms, academy reforms, auditing of these academies or auditing of universities... any guide book can be developed, but the system systematically sabotages it all"*[18].

As a result, the representatives of public organizations that defend science from the third sector do not always have enough power and resources to influence management decisions on the needs of science.

[15] Expert interview of O.Kolezhuk.
[16] URL: *https://docs.google.com/document/d/1VeqQarjD2NbYzJr6rd5G4k5jGkrNSeWC/edit?rtpof=true*.
[17] Expert interview of A. Senenko.
[18] Expert interview of A. Senenko.

Mechanisms for advocating science to public authorities during the war

For the most part, scientists use the advocacy tools and mechanisms that are the most understandable and accessible to them — letters for support (personal), petitions (together with science activists and supporters), recommendations (as part of working groups at institutions), and participate in public hearings. The latter tool has not shown its effectiveness yet, as organization of such events takes place without awareness of organizational processes, namely: announcements of hearings are published too late to prepare for them thoroughly, facilitation of hearings is carried out without special analytical preparation, mostly spontaneously, and the results of discussions are not always processed and published for review on time.

At the legislative level, there are a number of mechanisms that can be used to advocate for the needs of scientists to the authorities, including public councils, working groups, and strategizing.

Public councils and working groups

The MES has a number of councils dealing with scientific and technical activities. An important place among them should belong to the MES Public Council, but there are a lot of problematic moments in its work.

In 2023, the **MES Public Council** had not a single civil society institute (CSI) specializing in science. None of the Public Council members chose science as their preferred area of work. Science issues are not allocated to a separate subcommittee, but are combined with vocational and professional pre-higher education.

Representatives of CSIs complain about the difficulty of getting into the Public Council, as there is a list of CSIs approved by the MES that can delegate their representatives to the constituent assembly. It is almost impossible for a newcomer to get in, as it requires that the CSI already has a certain reputation and experience in the field of education and science.

Specialized councils within the MES play an active role in advocating the interests of the scientific community to the authorities, in particular:

- Council of Young Scientists;
- Council of Vice-Rectors for Research;
- Council of Vice-Rectors for International Relations;
- Council of Heads of Scientific Institutions.

In June-August 2023, the Board of the Council of Vice-Rectors for Scientific Work was involved in improving the conditions for conducting competitive selections of basic research, applied research, scientific and technical (experimental) developments of the MES under budget program 2201040 (main competition and competition for young scientists). Thanks to several strategic sessions the proposals from the Council were developed, which were taken into account by the MES when announcing competitions for 2024. In September 2022, the regulations on the Council of Vice-Rectors for Research and the Council of Heads of Scientific Institutions were updated[19].

The practice of establishing ad hoc working groups to develop certain regulations also continued. In particular, the development of a methodology for auditing scientific institutions and higher education institutions will require the creation of a working group that will include representatives of stakeholders[20].

[19] URL: *https://mon.gov.ua/ua/npa/pro-diyalnist-deyakih-doradchih-organiv-ministerstva-osviti-i-nauki-ukrayini-u-sferi-naukovoyi-naukovo-tehnichnoyi-ta-innovacijnoyi-diyalnosti-ta-viznannya-takimi-sho-vtratili*.
[20] Expert interview of Denys Kurbatov.

Public discussions

In spite of the full-scale invasion, public discussions of draft regulations in the field of scientific and technical activities did not stop. The only format of public discussions during this period was electronic consultations, and no other formats were used. On the one hand, this was related to the existing safety risks that limited the possibility of discussions in other formats. However, the practice of other organizations and initiatives shows that it is possible to use the in-person format of events even during martial law, provided that the necessary safety requirements are met.

The number of participants in the public discussions was small, with 1–3 participants on average. This number depended on the topic of discussion. In particular, the biggest number of participants joined the discussion of the draft resolution of the Cabinet of Ministers of Ukraine on improving the training of higher education applicants for Doctor of Philosophy degree, which was related to the significant resonance in the scientific community caused by this initiative. However, even such activity in this discussion did not allow involving all stakeholders, and the publication of the updated Procedure for the Preparation of Higher Education Applicants for Doctor of Philosophy and Doctor of Science Degrees in Higher Education Institutions (Research Institutions) in May 2023 sparked heated debate in the scientific community in social media.

The results were not uniform: the proposals of some stakeholders were mostly taken into account, while others were not, which was due to the different qualitative level of their preparation. Thus, it is clear that public discussions have not become fully functional as a tool for advocating the needs of science.

According to NAS President Anatolii Zahorodnyi, public discussions are a fairly simple and important mechanism for interaction between the government and public. However, it is necessary to ensure better organizational support for them. In his opinion, often not enough time is allocated for public discussions, or information about the start of a discussion of a document is not sufficiently disseminated to the public[21].

At the same time, it should be noted that public discussion as an advocacy tool has its own significant limitations. It is carried out at the final stage of development of legal acts, when the key concept of the document is formed and it is no longer possible to change it. Oleksiy Kolezhuk emphasizes this point: “The wrong “architecture” of the document and the wrong logic cannot be corrected by small amendments at the stage of public discussion.” In addition, the legislation does not provide for the possibility of suspending consideration of the project at the public discussion stage if a significant part of the scientific community opposes the innovations[22].

As the indicative calendar of public consultations of the MES is adopted and published before the beginning of each year, stakeholders can plan their advocacy activities for the year[23] ahead. Meanwhile, this is not obvious knowledge for those who can participate in public consultations, interagency communication about this is not sufficient, and public information is neither accessible nor understandable, as our respondents repeatedly highlighted. Nevertheless, according to the Ministry representatives, there were no deviations from the calendar in the year and a half of martial law (2022–2023).

[21] Expert interview of A. Zagorodny.
[22] Expert interview of O. Kolezhuk.
[23] URL: *https://mon.gov.ua/storage/app/media/rizne/2022/12/09/Oriyent.plan.MON.shchodo.prov.kons.z.hromadk.2023.pdf*.

Strategizing

The practice of strategizing, when experts are involved in the initial development of the conceptual framework of public policy in science, is not systematic in Ukraine. During the full-scale invasion, the situation has worsened as the possibilities for live communication are very limited. The very principles and methodology of strategic sessions are incomplete.

In 2023, when the military situation in Ukraine became more stable and significant personnel changes took place in the MES (the appointment of new Minister *Oksen Lisovyi* and most of his deputies), the Ministry initiated a broad engagement of the scientific community in strategizing on the ways to transform Ukrainian education and science. A total of 1,700 experts were involved in strategizing, including 549 in the field of science.

A strategic session of this scale was an important event in the history of Ukrainian science, but it did not meet the expectations of the scientific community. The strategic session highlighted the existing flaws in the methodology and format of the event, which were also affected by the martial law: the work was mainly done in an asynchronous format of providing recommendations in writing, there were almost no opportunities for live discussion (online or offline), and the role of moderators and facilitators in facilitating the discussion was minimal. There were also many questions about the quality of the proposals submitted: scientists sometimes demonstrated a lack of understanding of the legal framework for scientific and technical activities and peculiarities of public policy making.

Effectiveness of science advocacy to the authorities

The effectiveness of science advocacy to the authorities has always raised serious questions. As Oleksiy Kolezhuk notes, “in the last ten years, the interaction between the government and science has been reduced to interaction during the preparation of the budget for the next year (and it ended with this “skirmish”, until the next year).”

Executive authorities can ignore famous people who advocate for science, for example, offering to join the work on a draft law at the stages when it is already difficult to make amendments, but this is not always like this[24].

On the other hand, when interacting with the authorities, scientists notice when they are rejected. When everything works, the results of communication are at best reported as a fact or are kept silent. That us, the negative things are spoken about in public, while the positive things are forgotten.

A good example of cooperation between the authorities and public is direct interaction with active scientists-bloggers. In particular, participation in meetings, notification of planned events, open discussion of legislative and managerial initiatives, as well as effective procedures to influence the decisions important to the scientific community. *Anton Senenko* notes: *“The Scientific Committee always communicates and, if there is any interesting news, the Chairman of the Scientific Committee Oleksiy Kolezhuk draws attention to it (tags or sends a link to the messenger). Kyiv Academic University does the same”*[25].

The maximum influence of scientists on government decisions was exercised through proposals that were collectively developed and submitted for consideration[26]. Expertise

[24] Expert interview of A. Senenko.
[25] Expert interview of A. Senenko.
[26] Expert interview of A. Senenko.

of innovations related to organization and financing of scientific activities in the country by scientists should be transparent, understandable, and accessible.

The beginning of the full-scale invasion significantly slowed down the lawmaking activity, primarily at the level of the Verkhovna Rada. Consideration of a number of draft laws was postponed due to the priority of resolving security and defense issues. "Draft laws that have already been developed and need to be adopted have been postponed for an indefinite time. Some of them, thanks to the assistance of the relevant committee of the Verkhovna Rada, have been squeezed through in the form of "excerpts", with arguments that it is needed immediately," notes Oleksiy Kolezhuk[27]. Of course, this situation greatly complicated the process of advocacy for science at the parliamentary level.

Lawmaking activities within ministries and agencies also had to be adapted to the realities of the war: first of all, it was necessary to make a decision to stop cooperation with russia as an aggressor and its ally belarus. However,t implementation of the planned lawmaking tasks continued, including according to the operational plan of the MES[28].

Science advocacy to local authorities

Scientists have mixed opinions about the work of local governments, both positive and negative. An individual approach has different consequences. Some informants praised their city mayor, telling about their help and caring attitude. For example, *Olena Khadzhynova*, acting rector of Pryazovskyi State Technical University, noted that the deputy head of Mariupol City Council *"Serhii Zakharov helped to establish contacts with Donetsk Regional State Administration, draft letters and requests for support.* [NAUK07]. The heads of Kharkiv and Odesa Regional Military Administrations assisted the research institutions of the region in solving urgent organizational and economic problems.

Some respondents noted the complete inaction of the authorities, both local and central. For example, the management of Kherson library did not answer the phone calls of ordinary employees, and there was no one to receive instructions from because no one wanted to take responsibility in a situation of complete chaos and disorientation. Oksana Shestakova, the Head of the Department of Social and Cultural Activities at Oles Honchar Kherson Regional Universal Scientific Library, said that *"the local authorities fled, the governor disappeared on the first day, and there was only one mayor who called to remain calm..."* [NAUK08].

Education and healthcare were prioritized, while scientific and cultural heritage did not receive any assistance [NAUK03]. Request to local authorities for demining (on their own behalf and on behalf of the President of the National Academy of Sciences) did not yield results, as the priority of military administrations was to save lives, ensure security, and supply food. Mine clearance was carried out in stages, with priority given to residential areas [NAUK01].

The review of actors, mechanisms and tools of science advocacy to the authorities in this book is not exhaustive, but refers to those examples that have become relevant during the crisis. Observations show that there is a lack of awareness among scientists about the possibilities of advocating science to the authorities, and the lack of communication in the implementation of mechanisms for protecting, preserving and restoring scientific activity leads to misunderstandings and position conflicts.

[27] Expert interview of O. Kolezhuk.
[28] Expert interview of I. Taranov.

2.4. Science advocacy for business and public

The role of scientific results for business development is recognized by those entrepreneurs who consider it to be important to ensure high quality of their products and their company development. They understand that cooperation with scientists is a key element of the process of developing and implementing innovations. Scientists help to conduct a detailed analysis of scientific literature and the latest research, which allows them to improve business processes, find relevant topics and occupy new niches in the market, and introduce innovative technologies.

Nataliia Lopina, founder of ClinCaseQuest simulation-based training platform, summarizes the benefits of the business and science tandem:

- Scientists are capable of developing new technologies and methods that open up new business opportunities, help improve business processes and even take a leading position on the market.
- Scientists can bring in-depth knowledge and expertise in a particular field, which helps businesses solve complex problems and develop effective strategies.
- Collaboration with scientists can provide businesses with access to various resources, such as laboratories, equipment, databases, or other infrastructure.
- Scientists can conduct validation studies and analyze them, which helps businesses make informed decisions and form effective development strategies [BUS03].

According to the managers of commercial organizations, businesses "do science for money" and make money from research. According to *Yuriy Rebets*, Scientific Director of Explogen LLC, *"We do our own science and are not particularly interested in external developments. We have the tools and staff to solve scientific problems within the company. The purpose of our projects is scientific developments"* [BUS01].

Entrepreneurs assess risks in advance, get to know their competitors and calculate costs and benefits. *"Ukrainian business in Ukraine is focused on quick solutions, on creating a product (technology) that can pay off investments very quickly. That is, the period of investment and expectation that these investments will return is shorter than in countries with a stable economic situation. That's why business is very hesitant to look towards science. Science is perceived as something that takes a very long time, where you invest money but don't get a concrete result. Science is perceived as a field that has great risks,"* says Maria Lemberg, Director of Fedir Shpyg Charitable Foundation [BUS02].

The peculiarity of any business is its focus on getting profit. However, as Maria Lemberg observes, *"some Ukrainian companies understand that they can get an important product from scientific institutions, some do not."* [BUS01] If research is easier and cheaper to conduct in Ukraine, then international partners will be interested in cooperation. It is not economically feasible to develop a scientific infrastructure from scratch. Therefore, it is sometimes easier for foreign companies to outsource research to other firms than to conduct it themselves [BUS01].

An important advantage is the presence of good investors (who accept the risks of not achieving a successful outcome), a solid team, modern equipment and own technologies to ensure the success of future products [BUS01].

A business representative mentioned finding competent staff as a serious problem. They were able to find highly qualified specialists through personal acquaintances and recommendations of recognized authorities. There are a lot of proactive scientists with interesting ideas in Ukraine, but business structures are cautious and carefully select ideas. For example, Yuriy Rebets says: *"Even the best ideas don't always work. We, as a*

company, have to think about the financial side of things, assess risks, think about whether we can make a profit or not" [VUS01]. The ability to present your research project is becoming a key factor in successful interaction and entry into market relations.

Cooperation of businesses with partners in the field of science takes various forms, for example, solving a social problem related to health restoration and rehabilitation after chronic stress and trauma, and implementing online educational projects for schoolchildren. Close, direct conversations with scientists highlighted the problems scientists have and helped us delve deeper into their solutions. The goal of science-based businesses is to bring Ukrainian science to the level of those enterprises that independently generate their own innovative products. Unfortunately, today Ukrainian scientists are relied on to carry out outsourcing projects for global companies. Business understands how important it is to identify promising research areas and provide mechanisms for implementing scientific results and developments [BUS02].

According to *Maria Lemberg*, entrepreneurs have little understanding of how collaboration with scientists can help them implement innovative projects. First, they are afraid of legislative barriers: *"There is no transparent and understandable mechanism for collaboration with scientists. This is also related to legislative norms that complicate this process (I mean some rules on intellectual property, patenting, inability to work directly with scientists, monopolization of science by the state, etc"* [BUS02]. Secondly, economic and institutional constraints that make scientific innovations require long-term investments with no certain chance of profit stop them: *"In Ukraine, science can only be done when scientists belong to a certain institution. In other words, companies can theoretically finance the creation of research centers, but it is expensive and it is not clear when it will pay off. Only very resourceful companies can create and finance the work of research centers"* [BUS02]. Thirdly, scientists do not participate in the market presentation of their ideas, and there is no marketing strategy to integrate scientific developments into any business processes. *"There are many reasons why businesses should invest in science and cooperate with scientists. But these reasons need to be explained for a long time. This issue is undercommunicated. And it is the scientists themselves who should communicate with business in the first place (explain why it is important and profitable)"* [BUS02].

Explogen LLC is a good example of business cooperation with science. The company management decided to participate in the educational process. They invite talented students and young scientists to their laboratories, and they carry out research for their diploma and other papers. In addition to internships, they organize weekend schools where students from different regions work for free in a modern laboratory and learn new technologies. For the past two years, the company has been running a lecture course and practical classes together with the university. The course is attractive because it is funded by DAAD and there are scholarships for students [BUS01].

During crises, it is a difficult and balanced decision to support knowledge-intensive projects that work to overcome the catastrophic situation and negative consequences. Grant programs for knowledge-intensive businesses are becoming a useful practice: *"We started active cooperation with scientists only in March 2023 when we received a development grant as a dual-purpose project important for the country's defense capability and post-war recovery from the Ukrainian Startup Fund with the support of the Ministry of Education and Science of Ukraine,"* says Nataliia Lopina [BUS03].

As part of this government support, the organization was able to hold tenders to select scientists and experts to further implement the planned tasks, build a strong team of specialists to implement the grant, and pay taxes to develop the Ukrainian economy. *"We have reinvested almost 100% of the grant funds in scientists and medical experts"* [BUS03].

To do this, scientists had to look for new formats for transforming their experience and ways to monetize scientific expertise, and the business structure became a platform for support and further development.

Although regional higher education institutions had close ties with local businesses, during the war period enterprises themselves appeared in a critical situation, so it made no sense to rely on them [NAUK07]. Scientists were able to reach business structures through their graduates who work at the relevant enterprises or are their owners. This is exactly what happened at Sumy National Agrarian University. As SNAU Vice-Rector Yuriy Danko stressed, "the western model of cooperation with alumni worked completely!" [NAUK04].

The subject matters of the requests are not only related to scientific purposes. For example, in addition to requests for corn seeds for experimental sowing, an agricultural university asked alumni for financial assistance to pay salaries to university staff, and received support [NAUK04].

Summing up the review of opportunities for advocating science to business, the good news is that funding for science is based on models of interaction with four types of stakeholders — patrons, donors, sponsors, and volunteers. According to the contribution of stakeholders to the common cause of preserving, restoring and developing science, the following types of stakeholders are distinguished:

A patron seeks to remain in history, so the payment for their contribution may be a mention and honoring with a plaque, a lecture hall in their name, etc. Interaction with the patron ensures recognition among other patrons, both for the patron and the institution they have supported.

A donor makes a contribution because it is beneficial for them to see a project develop as it ensures sustainable development of society through the creation of new jobs, spread of value narratives about the attitude to people from certain vulnerable groups in society, etc. The donor ensures that the organization fulfils its social mission of shaping values in line with the society needs.

A sponsor is someone who expects a return on their investment. An ordinary person wants to be involved with those who have sponsors — people who earn enough to "share" part of their income for a common cause.

A volunteer is someone who shares values and is ready to invest their time in improving the situation of representatives of a particular vulnerable group in society.

For a stakeholder it is important what status they will have if they are persuaded to change their attitude to science and share their resources in the interests of scientists to support their research.

Scientists have been able to involve NGOs and people who care to solve their problems. For example, *Yuri Logvinov*, Acting Director of O.Ya. Usikov Institute of Radiophysics and Electronics, says this: *"Volunteers, employees and simply caring people have become the basis for the survival of the institute"* [NAUK02]. The most effective channels for preserving cultural heritage were those of volunteers and the public sector, including international ones.

As we can see, the interaction of business with scientists provides win-win opportunities: scientists, getting opportunities for research and conducting classes with students at commercial institutions share research results and talk about business opportunities with other partners in their networks which helps to diversify economic risks and raise awareness of business services. Meanwhile, public opinion formation is important for science advocacy to business, which requires interaction with mass media and understanding of media logic and social media logic[29].

[29] van Dijk, T.A. 1996. Discourse, power and access. In Texts and Practices: Readings in Critical Discourse Analysis, eds. C.R. Caldas-Coulthard and M. Coulthard, 84–104. London: Routledge .

2.5. Science advocacy in cooperation with mass media and other media

French positivist sociologist Emile Durkheim pointed out the prejudice that "science is an antagonist to public opinion, with which it fights and corrects its errors". He emphasized that "science can only succeed in society if it has sufficient authority, which it can only have in public opinion"[30].

The media, in particular the mass media, should be seen as an advocacy tool to government, business and public. Not in the least, the credibility of scientists is formed by whether people see them in the media. The mass media represent society through a limited number of individuals — politicians, celebrities and experts on "all matters" who are more active than real scientists. When people are informed about the issue, understand what it is about, they can share the values of science, and this support will help achieve advocacy goals. That is why, before launching an advocacy campaign, it is important to find out what people other than scientists know about the needs of science.

It is difficult to overestimate the role of the mass media in science advocacy, but does society use the mass media's communication resource to the fullest? Interviews with experts show that there are lots of barriers to interaction between scientists and journalists. The first barrier is the complexity of scientific knowledge. For people to understand what scientists do and why it is important, they need to talk about science in the simplest and most interesting way possible. Journalists do not have enough time (and sometimes experience) to delve into a topic and present it in a quality manner. Scientists also don't always have the resources or interest in editorial work. Another barrier is the stereotype that scientific news stories are not interesting to the public (and will not be covered).

The experience of the leading agency (Ukrinform) shows that news about the discoveries of Ukrainian scientists is interesting for news services, and are actively disseminated and discussed by information consumers [MEDIA04]. Editors of news agencies note that the choice of topics for publications is based on the challenges of reality (destruction during the war, suspension of the educational process, loss of life, etc.), but it is not always possible to get a comment from a scientific expert promptly as it requires a number of administrative approvals and substantive clarifications: each time it is a quest.

According to *Maryna Synhaivska*, Deputy Director General of the Ukrainian National News Agency Ukrinform, in order to get a comment from a scientist abroad, journalists had to "*prepare questions, send them through the head office to get permission to answer our questions, the scientist answered, the office approved it, and only then we printed it*" MEDIA04]. The editors emphasize that it would be good if there was an up-to-date database of scientists who could be contacted promptly on various issues as it is the presence of scientific experts that draws the authorities' attention to scientific expertise [MEDIA04].

Thus, we consider advocacy as a special case of communication activities. Advocacy of Ukrainian science is, to a large extent, an intervention in the environment of ignorance about the needs of scientists, misunderstanding of the problem, its roots and consequences, or unwillingness to solve it. The purpose of this

[30] Durkheim E. The Elementary Forms of Religious Life. Zhupansky Publishing House, 2002, p. 198.

intervention may be to inform, persuade, and shape the culture for further preservation, restoration, and sustainable development of science.

Media activity plays an important role in science advocacy through highlighting problems and finding potential donors. A well-written article arouses the interest of a diverse audience, which increases the chances of finding the right partners and sponsors. If scientists actively and regularly talked about their needs and problems in social media, journalists noticed these posts and prepared a separate article [NAUK06]. They got in touch with the contact person, arranged an interview, recorded comments, etc. The effectiveness of media communication depends on the scientist's openness and willingness to become a public speaker and disseminate messages to a mass audience. This activity helps get a decent result: to raise money, disseminate useful information, and show effective examples of solving crisis problems. This is exactly what the scientists at the Institute of Radio Astronomy did. *"We raised 10-15 thousand euros, which is not a lot of money, but it helped us a lot. We want to spend it on the restoration of the telescope, we are waiting for the demining to be completed,"* said *Vyacheslav Zakharenko*, Director of the Institute [NAUK01]. *Ivan Moisiienko*, a lecturer at Kherson University, said the same thing: *"The media helped draw attention to the problem. The dissemination of information in the mass media was very useful: people learned about our experience and asked for advice"* [NAUK06].

The business expert emphasizes the importance of a deliberate and balanced information campaign developed by communication specialists and not limited to social media. *"We have a person in charge of partnerships who communicates with potential partners. We really want communications to go beyond the Facebook page"* [BUS02].

Another expert in the field of knowledge-intensive business sees the relationship of science with mass media as purely about popularizing scientific developments. The content of their social media is limited to news about the achievements and development prospects of an innovative company. In their opinion, posting spontaneous posts is sufficient to keep in touch with international and domestic media [BUS03]. Thus, the absence of representatives from the organization to communicate with the media is not only related to a lack of financial resources, but also to a lack of understanding of the principles of mass communication and diversity of reactions of the media audience.

Representatives of international organizations get acquainted with Ukrainian science also through local media. In particular, the experts mentioned the section on education and science on the Ukrainska Pravda portal, popular science media Kunsht, and other media [INTERNP01].

The initiative sometimes comes from journalists, who turn to scientists for information and comment [NAUK05]. As a rule, scientists find it inconvenient to ask for help in solving their problems [NAUK02]. This mental barrier actually negates the use of the few advocacy tools available to the scientific community. At the same time, scientists recognize that it is impossible to solve a problem without drawing attention to it. In fact, these scientists do not use various opportunities to articulate their needs. One expert noted that *"we have never managed to raise funds and find new partners through publications, probably because publications in specialized scientific and technical journals are more effective in this area"* [NAUK02].

The unsystematic nature (chaotic and without specific goals) of information activities does not always contribute to the rational allocation of resources and time to solve communication problems. In one way or another, scientists with an active civic position manage to make themselves known in the public space and draw attention to specific problems. *"We did*

not aim to solve any specific problems with the help of publications in the mass media. We drafted appeals and petitions and sent them to various mass media outlets. In other words, we were trying to keep the information front open," said *Yuriy Danko* [NAUK04].

The media also join to raise awareness of the problems that arise on an individual level. *"There was a situation when a colleague who had moved to the territory under Ukrainian control was asked by the institution to 'vacate her room'. I wrote about it on social media and contacted journalists. The media in Ivano-Frankivsk region wrote about this case, and the institution that provided me with shelter allowed me to stay for a while longer." Oksana Shestakova*, an employee of Oles Honchar Kherson Regional Universal Scientific Library, shared this case [NAUK08]. Although this example does not provide a systemic solution, this media material helped in a specific situation and suggested to others what to do in similar cases.

At the same time, building regular relations with the media does not only require knowledge and skills, but also regular work with the media. *"In peacetime, we communicated a lot with the media, both local and national (in particular, Inter and 1+1). I have contacts of these journalists and when I need something, I can write to them,"* said Oksana Shestakova [NAUK08].

There is no time to develop long-term relationships during the crisis, so if there was no pre-established practice of interaction with journalists, we had to rely on mutual movement towards each other. *"We didn't have time to look for someone specifically. Journalists found us on their own. ... In these [media] publications, we conveyed the message to the world: we are alive, we are recovering, we are working"* [NAUK07].

During crisis, communication with media representatives becomes more cautious as the speaker needs to anticipate short-term and long-term consequences of public disclosure of certain facts related to scientific activities. *"Our main problems, no matter how acute, cannot be discussed openly through the media and social networks today,"* notes one expert [NAUK03].

Initiative heads of departments were able to interest and unite civil society organizations to address priority issues. They had to be active in many unusual formats in the context of hostilities: giving interviews to Ukrainian and foreign mass media, participating in live broadcasts, speaking at conferences and online meetings. Informants note a decline in the level of communication and intensity and culture of discussions on public platforms: *"There are results, but now there is a decrease in communication on various previously active platforms"* [NAUK03].

Using the potential of social media and discussing science in society

In Ukraine, social media sometimes replace official sources and media for communication and representation. They do not only reflect public opinion, but also form it. The intensive use of social media by scientists and activists to advocate for their needs is a sign that official sources of information are unavailable or simply inconvenient.

Knowledge-intensive businesses understand how to use social media such as FB and LinkedIN, for example, and reach international audiences, while scientists and scientific institutions use social media as a platform for everyday communication. It is worth either learning how to communicate or interacting to reinforce each other. Businesses need knowledge for innovation, and science needs tools for social interaction.

Social networks are a means of personal communication, and they should be seen as auxiliary for positioning and building reputation. For international interaction, official portals, websites, platforms, as well

as stationary offline platforms, which are actually museums, libraries, laboratories, and institutes, play an important role.

Communication on social media is too emotional. One of the respondents shared that he was very upset by criticism on social media. Advocacy needs to take into account the risks as it is not uncommon for those who voice the problem to be criticized, rather than those who prevent it from being solved. Communication during implementation of advocacy actions can be aggressive, with unexpected reactions. Awareness raising (trainings, seminars) to develop critical thinking and increase media literacy will help prepare for crisis situations during advocacy campaigns.

The monitoring of social media on discissions of science on social media and online resources during the first year of the war (analysis of the array by Youscan online data collection software) allowed analyzing the discourse of science in the first year of russia's full-scale invasion of Ukraine. The analysis revealed communication gaps, which, if not taken into account when implementing advocacy activities, lead to errors in the expected results. Communication about the needs of science produces mainly negative discourse, while positive discussion of science is mostly related to the educational process rather than purely science, such as advertising for admission to higher education institutions or feedback on the quality of tutoring services. Opinion leaders are active on social media both in advocating for science and in popularizing it.

Although the monitoring was conducted based on specific mentions of science, the topic of educational process was one of the most frequently discussed topics in the discourse of science on social media throughout the year. In spite of the fact that information about the activities of scientific schools, research areas, and scientific results could help to advocate for the needs and promote certain educational institutions during admission campaigns, the educational process was communicated one-sidedly, purely functionally (in March — resumption of studies, in June — the cancellation of external testing and the state exams, in August-September — admission, then awards, celebrations, and writing diploma papers). Where information about science was expected, empty advertising messages prevailed. The topic of celebrating holidays and memorable dates takes a significant share. People send greetings on holidays and dedicate events to historical events and great figures.

The tone of discussion of advocacy for the needs of science during the war was mostly negative if the topics were related to the consequences of the war — destruction, loss of life and attempts to spread information about the true history of Ukraine. For example, in Facebook group "News of Higher Education", the discussion is mostly negative as each post has a provocative intent and stimulates a heated discussion, often not on the merits (for example, the post: "In the near future, the funds are planned to be reallocate (the Government's decision) within the MES for the Museum of Science", first two comments: "All that's left to do is to put science in a museum", "It's better to make a museum of the 95th quarter"). This can be both a stylistic device to ensure feedback support and a lack of communication culture. This style enhances effectiveness of interaction and even shows itself as a sustainable model. At the same time, it does not contribute to the formation of a constructive discussion on reforms and support for existing mechanisms. The group also disseminates news from online media resources with national coverage (Suspilne, FREEDOM, EspressoTV, Channel 5, NV, Apostrophe TV, Ievropeiska Pravda, etc.), messengers of regional broadcasters — Visti Prykarpattya, Novyny Zhytomyrshchyny, D1, Halych Info, etc.). This should be used in further planning of interaction with the media and other media professionals (Report).

The topic of interaction of scientists with the army and discussion of the educational process is positive, but the posts in this topic

are monotonous. Accordingly, advocacy for funding research aimed at the defense sector would be more effective and supported by society. By the way, the information on admission does not include either scientific specifics of the institutions or benefits of researching scientific topics, scientists are not often mentioned. The posts on advocacy have a characteristic emotional distribution — most of them evoke negative emotions. The authors of the posts express distrust of advocacy tools (public discussions) or disbelief that they can achieve their goals by advocating for certain needs. This should be taken into account in future activities as a risk because any useful ideas may be perceived by the public as false due to prejudice against advocacy in general. The topic of academic dishonesty is discussed separately, with different assessments, including positive ones — there are proposals to have "qualified specialists" write papers for students who are submitted with an emotion of care.

Social media reflects reality, "but it's not for sure". In the topic of science, communication is mostly ritualistic and creates "white noise" that prevents dissemination of more important information. The emotion accompanying this topic is joy or respect. For example, there is a Christmas greeting from the Department of Biotechnology. This would be appropriate if the department had been active in informing about its scientific or educational activities during the year, but no such messages are observed. The awarding ritual has a significant number of mentions. More interaction in the discussion of science in social media is observed with posts from such areas as education, history, culture, applied analytics, medicine, psychology, and religion. These are the areas where the authors of messages most often appeal to emotions or manipulate the needs, fears, and "pains" using advertising techniques. Pseudoscientific information is also popular among social media users. For example, there is a post where people rejoice that "a Ukrainian scientist has proved that prayer changes blood and really heals" (in the original text the words are written in capital letters). In general, the scientist is presented in social media messages as an archetype of an alchemist, a magician — a person who can invent a panacea. No one knows how he or she does it. People believe a self-declared scientist because he or she calls himself or herself an expert.

The leaders in discussions of science during the war were Kyiv and region, Lviv, Kharkiv, and Dnipro regions. On the one hand, these cities became hubs during the war. On the other hand, they are the cities with the most developed scientific communication, which does not facilitate decentralization of information about science. Communication with other scientists or stakeholders was carried out on an ad hoc and unsystematic basis: posts on social media, videos, photos are not a planned conscious activity, but rather occur spontaneously, "on the inspiration" of the scientists or supporters themselves. Meanwhile, dissemination of information with involvement of the community (volunteers, friends, neighbors, etc.) has yielded results — a video was created, a post was published on social media, which attracted the attention of the media and benefactors (a photo of the destroyed building was taken, posted it on social media, and funds were raised). Such communication should be facilitated by creating the environment for science communication, using the spaces of open laboratories, museum and library hubs, science parks, etc.

Chapter 3. Science advocacy to international stakeholders

International advocacy for Ukrainian science began to develop since the full-scale invasion. Before science diplomacy in Ukraine was based on models developed in the Soviet context, when Ukrainian science always remained in the shadow of dominant Russian science. In fact, Ukrainian science diplomacy was weak, and before the war it was often implemented together with Russian science [MEDIA02]. During the first year of the full-scale war, many issues in the advocacy of Ukrainian science in the world have moved forward, but it will take time to build a sustainable model.

International stakeholder groups

Several general categories can be identified: governmental and supranational institutions, non-profit NGOs, universities and other scientific organizations, scientific journals, businesses, and individuals. The main stakeholders include the EU Commission and its Directorate-General for Research and Innovation (*DG RTD*). Horizon Europe project is one of the levers of assistance and influence of the EU Commission on the scientific situation in Ukraine. The EU governments and their respective ministries also belong here. On the US side, these are the National Academy of Sciences and the National Science Foundation (*NSF*). Non-profit non-governmental institutions include scientific societies, such as the International Science Council (*ISC*), the World Academy of Sciences, *TWAS*, and the American Association for the Advancement of Science (*AAAS*). An example of a private charitable foundation is *Simons Foundation*, and an international non-profit organization is *Aspen Institute*. Universities, scientific societies, and individuals (individual scientists or interested scientific groups) also play an important role in supporting Ukrainian science.

Jan Marco Müller, Coordinator of Science Diplomacy at the EU Commission, noted that the war showed the "dark side" of science diplomacy. It has shifted the perception of its image from the romantic "keeping channels of communication open when diplomats are silent" to the restriction of scientific contacts and projects, which, for example, the USA has introduced against Russian scientific institutions. Or the exclusion of Russian

institutions from the Horizon Europe. Müller also refers to the need to help refugee scientists as the "dark side" of science diplomacy and stresses the need to revise the approach to science diplomacy during and after this crisis. According to him, scientists themselves are currently taking diplomatic action, for example, by refusing to cooperate at the individual or research group/institutional level with Russian institutions whose rectors have signed a letter supporting the war.

Science diplomacy is seen as application of science, technology and innovation in three policy dimensions — participation in foreign policy making through scientific advice (science in diplomacy), facilitating international scientific cooperation (diplomacy for science) and using scientific cooperation to improve international relations between countries (science for diplomacy)[31]. TWAS expert (The World Academy of Sciences) Peter McGrath adds one more dimension — diplomacy in science, citing examples of the relocation of scientists during military crises (war in Ukraine, Syrian crisis, military coup in Myanmar).

International organizations are an important actor in supporting and developing Ukrainian science during the crisis. Unfortunately, their resources are not fully used, and there are not enough programs to help and support Ukrainian scientists. The reasons for this are a lack of institutional flexibility of the organizations themselves and a lack of understanding of the realities of Ukrainian science. At the beginning of the Russian invasion, European organizations (International Science Council, InterAcademy Network, relevant bodies of the European Commission) were much better informed about this issue than American ones.

Lots of international organizations have responded to the call for assistance to Ukrainian scientists. Among the assistance provided by international organizations, respondents most often mentioned the Simons Foundation's support programs for scientists (USA) (11 mentions), DAAD (7 mentions), Erasmus+ (3 mentions), scholarship from Max Planck Institute for the History of Science (3 mentions), Krzysztof Skubiszewski Foundation (3 mentions); UNICEF (2 mentions); Scholar at risks (2 mentions); Scholarship (2 mentions); ALLEA-EFDS (2 mentions), and many others (see the list below). One respondent was able to receive several international grants at once: "I have won 10 international grants (USA — 5, Austria — 2, Vietnam — 1, Italy — 1, France — 1)". However, it is much more common for a scientist to find it difficult to get help for various reasons [Lutsenko A. et al (Third Wave Report)].

The institutional flexibility of international organizations made it possible to launch direct support programs for Ukrainian scientists who remained in Ukraine. Non-resident programs were characterized by loyal bureaucratic procedures. However, a **lack of understanding of the realities** of Ukrainian science can be seen in situations where communication takes place through structural units that are not competent promoters of science. In expert interviews, scientists sometimes mention auxiliary administrative units in relation to international cooperation. This may indicate that such auxiliary administrative units do not exist or are ineffective (so-called international departments, etc.).

Unfortunately, a small number of Ukrainian scientists were able to take advantage of offers from international organizations to resume and continue their research activities in the form of grants and scholarships for research, business trips, participation in conferences and scientific internships,

[31] URL: *https://www.aaas.org/sites/default/files/New_Frontiers.pdf*. P. 5–6.

purchase of equipment, and one-time assistance.

In addition to grants and scholarships for scientific activities, respondents often mentioned one-time or regular humanitarian material assistance (food or money) from charitable organizations (38 mentions). It should be noted that only one of the scientists mentioned salary as an aid, i.e., he perceives his fee as unexpected and optional (accidental) assistance, rather than a well-deserved reward for the amount of work performed. The payments from the UN were mentioned most often, and less frequently — from the International Committee of the Red Cross (ICRC) and the International Organization for Migration (IOM). Among the Ukrainian organizations providing humanitarian aid, the following were mentioned: Liudyna Charitable Foundation, E-Pidrymka, and E-Dopomoga [Lutsenko A. et al (Third Wave Report)].

Channels of communication with international partners

In expert interviews, scientists repeatedly noted that international contacts based mainly on personal communication have become the most effective. Face-to-face communication plays a key role in advocacy, but it is a significant burden for scientists in a crisis situation. Scientists delegate organizational tasks to specialists, but only when systematic work has been established before the crisis. Foreign stakeholders who had communicated with Ukrainians at the time of the large-scale war (either personally or through working groups) were better aware of the problem and acted more in line with the real needs of Ukrainian scientists. According to Cathy Campbell, it was the presence of Ukrainians at the working meetings (formal and informal) of the US National Academy of Sciences that allowed American stakeholders to quickly get up to speed on the problems that Ukrainian science faced. These meetings were attended by both institutional representatives (Olga Polotska from the National Research Foundation of Ukraine (NRFU), Oleksiy Kolezhuk from the National Council for Science and Technology Development) and individuals with extensive expertise (Yulia Bezvershenko).

Maud Fischet also speaks about the activity of the NRFU. At DG RTD (the European Commission's Directorate-General for Research and Innovation) had internal meetings where Ukrainians were periodically invited. They were attended by representatives of the MES and NRFU. These meetings were organized on a bottom-up basis, when interested experts familiar with the situation in Ukraine put forward their proposals for assistance. According to Peter McGrath (TWAS), two Ukrainians cooperating with the International Science Council (*IRC*) attended the inter-academic stakeholder meetings in the first weeks of the invasion. This fact, as well as the activity of representatives of the Polish Academy of Sciences, helped to "ground" the stakeholders in terms of understanding the real problems. Fishet attributes the effectiveness of such meetings to the impact of the COVID-19 pandemic, which has made people used to a hybrid (offline and online) format of meetings.

On the negative side, she notes the fragmentation of the Ukrainian scientific landscape: *"You have many strong actors who hate each other"*. This prevents creation of a common position from the stakeholders and makes it difficult for the European Commission to communicate with them, as one cannot be sure that an agreement on an issue will not cause a sharp negative reaction or opposition from other actors.

[10] Boundary organizations – organizations mandated to act as intermediaries between the arenas of science and policy (цит. за: Guston, D. H. (1999). Social Studies Sci. 29, 87–112).

Maud Fischet mentions another communication channel — *ERA4Ukraine* page which is an aggregator of initiatives to support Ukrainian researchers. In the first months after its launch, the page was visited by more than 100,000 users. Katherine Younger also mentions Twitter (X after rebranding) as a channel for communicating the Ukrainian agenda to stakeholders.

The majority of respondents name communication with fellow Ukrainians as the main source of information about the security situation in the country, including the situation in science. In addition, they also mention the major media (BBC, CNN, New York Times, Washington Post, The Guardian, Al Jazeera) as a source of information about the general situation in Ukraine, and scientific journals Nature and Science about the situation in the scientific field. However, scientific realities are hardly reflected in these general political media. Nature and Science, in addition to editorials about the war and the state of science in Ukraine, provide an information platform for Ukrainians. In 2023, Richard Stone came to Ukraine to make a story about scientists. Involvement of such journalists is a good tool for communicating Ukrainian realities to the international scientific community.

Establishment of international relations by Ukrainian scientists

Those Ukrainian scientists who had direct contacts with foreign colleagues used them to spread information about the war in academic circles and to organize help for themselves or other affected Ukrainians. An example was given by Alexander Kordyuk, Director of Kyiv Academic University (KAU). KAU had a joint project *UKRATOP* with German institute IFW Dresden. The essence of the project is the "brain circulation" in the form of internships for masters and PhD students in Germany with their return to Ukraine to defend their diploma/PhD thesis. Thanks to these contacts, it was possible to organize temporary positions for researchers who needed help in the first months of the Russian invasion. Direct contacts with foreign researchers also provide an opportunity to apply for grants that require international cooperation. In 2022, Horizon Europe enabled Ukrainian groups to join existing grants. The pre-war cooperation of KAU with IFW Dresden allowed scientists to apply for several international grants, including Eurizon, a project to create an excellence center — a Ukrainian-German laboratory from the German Federal Ministry for Education and Research (BMBF). The KAU Grant Office is a tool to help scientists fill out grant applications and find international cooperation.

According to Kordyuk, the fact that some scientists were forced to leave Ukraine and find temporary positions in the EU will help to establish pointed international contacts that will form the basis for further development of Ukrainian science.

Another recommendation is to attract leading international scientists to work in Ukraine. This could be an organized lecture at a university or a scientific school where foreign scientists are invited. Thanks to previous personal acquaintances, cooperation between Sumy State University and Yury Gogotsi began. Gogotsi is one of the most cited scientists in the world, an outstanding organizer of science, and working with him is beneficial for Sumy State University in terms of developing local science, attracting grants, international contacts, and increasing the university's visibility. Another positive example is the regular guest lectures of foreign scientists under the Visiting Professor program at Karazin University in 2023.

[11] Як стати цифровим інноваційним хабом: долучайтеся до вебінарів від європейських колег / Міністерство цифрової трансформації України. Урядовий портал: офіційна вебсторінка. 10 жовтня 2023 р. URL: *https://www.kmu.gov.ua/news/yak-staty-tsyfrovym-innovatsiinym-khabom-doluchaitesia-do-vebinariv-vid-ievropeiskykh-koleh*.

Ukrainians' influence on the rules of international grants

Institutional influence is the most powerful tool to influence the nature of international grants. The project experts Jan Marco Müller and Maud Fischet spoke about this using the example of Horizon Europe. Ukraine is an associate member of Horizon Europe and has privileged access to the decisions of the program committee. Participating countries delegate one representative to each thematic committee, where proposals for the rules of grant calls are put forward. The ideal case of influence is when a Ukrainian representative has a good connection with his or her committee and, on the other hand, with an institution in Ukraine, such as the Ministry of Education and Science, the National Academy of Sciences, or the National Research Foundation, which organizes discussions within the scientific community and provides proposals for calls. This is the golden rule of community advocacy at Horizon Europe.

Countering Russian influence in the international arena

Historically, Russia has invested significant amounts of money in promoting its interests internationally, and science is no exception. At the time of the full-scale invasion, russia was a member of global collaborations such as CERN, and its universities had partnerships with leading global universities, such as Skoltech with Massachusetts Institute of Technology. After the invasion, some of these ties were cut at the initiative of foreign institutions. Russia's status in CERN was frozen, the country was excluded from Horizon Europe, foreign universities cut ties with Russian partners, and scientific publishers imposed strict rules on the publication of articles by Russians (a ban on accepting articles with the participation of Russian universities, a ban on publications from universities located in the occupied territories and having Russian affiliations). These steps were prompted by the shock of the invasion, pressure from Ukraine and its partners, and the actions of Russian scientists themselves. Several experts pointed out that they were negatively affected by the open *letter* from the rectors of Russian universities supporting the war.

When they realized that they needed to influence the situation more intensively, Ukrainian scientists began to use social media: they sent letters to colleagues from other countries, organized blocking of Russian resources, and published appeals to their Russian colleagues. Some scientists made attempts to enter into a dialogue with their Russian colleagues. For example, Ivan Moysiyenko, Head of the Department of Botany at Kherson State University, said during an expert interview: *"I wrote an appeal to my Russian colleagues (the mailing list included more than 200 contacts). Most of them remained silent, but two dozen responded. Mostly, these were angry responses about the "8 years of the Donbas bombing" (typical Russian propaganda message). Professors, doctors of sciences, deans assured us that in a few days their army would capture Kyiv and everything would be over. It was horrible to read. I took screenshots of these letters and sent them to my colleagues abroad. (Although there were also one or two letters of apology)"* [NAUK06].

Ukrainians have launched several petitions for sanctions against Russian science. An open *letter* published in March 2022 was signed by almost 10,000 scientists from different countries, mostly Ukrainians, including those from the diaspora. The letter calls on the international community to block Russia and Belarus from accessing scientometric databases, exclude these countries from academic mobility programs, and boycott Russian attempts to host international events on their territory. The author of the letter suggested that these sanctions should not be applied to individual scientists or institutions that have proven their active opposition to the war against Ukraine. In his

opinion, such measures will force the Russian scientific community to actively oppose the actions of the authorities. Another open *letter* was published on the NRFU website in February 2023. As of October, it had been signed by almost 2.500 researchers from different countries. Its main statement is that "in times of war, science also becomes a weapon". With this statement, the authors of the letter are trying to counteract the widespread misconception that science is outside politics, which Russians use to fight off sanctions and accusations of working for the war machine. The letter calls for isolation of the Russian scientific sector — not to accept articles affiliated with Russian institutions for publication, not to sell scientific equipment, and to ban access to international scientific infrastructure for Russian institutions.

The open letter refers only to Russian scientific institutions and their affiliates, and not to scientists of Russian origin or citizenship in general. However, Russian scientists, both pro-government and pro-war, as well as opposition-minded, took the letter negatively and manipulated it by the fact that it allegedly called for refusing to cooperate with any scientists of Russian origin. Their reaction showed that in reality there are not as many differences between these two groups as it might seem. An example is several publications in T-invariant (online-media created by scientists and science journalists of Russian origin who were in opposition to the Russian government after the start of Russia's full-scale war), whose authors try to justify the need to remove Russian science from sanctions by appealing to the same principle of "science beyond politics". Such behavior is typical of Russians, even opposition-minded ones, from all sectors of culture or economy, who try to blur responsibility, remove part of the Russian state machine from the sanctions blow, falsely appealing to humanitarian values. Such actions only contribute to the impunity of the Russian Federation. Similarly, pro-government and pro-war Russian scientists act, who regularly publish open letters and petitions (sometimes co-authored with friendly foreign scientists) demanding that sanctions be lifted from science in general, from certain scientific fields (such as Arctic studies) and from universities such as Skolkovo Institute of Science and Technology (Skoltech) or Moscow Institute of Physics and Technology (MIPT), which are an important part of the Russian military-industrial complex, working to create new weapons, strengthen Russia's engineering and military potential. Or to create Russia's image as a progressive scientific power.

As noted in the section on science diplomacy, the crisis caused by the Russian war has shown its "dark side". Science is part of a country's positive image. Promoting Russia's scientific achievements whitewashes its image as a country, showing that it is on a par with civilized nations. The Russians use science, which is supposed to build peace, either directly for war or for PR, blurring their image as an aggressor. The prominent Ukrainian scientist *Ahatanhel Krymsky* (1871–1942), who was arrested without any reason and later murdered in prison by the Soviet totalitarian authorities, wrote about this very aptly: *"Every scientist or poet who finds the strength to work sincerely under our despotic system and does not go to protest and fight against that system, he simply sanctifies despotism and cements it with his conscientious work. For even though such a scientist or poet is silent and works quietly away from the world, his or her silent work seems to shout loudly to the whole world: "The modern system is not bad at all! After all, look, science, poetry, and cultural work are developing and flourishing under the modern system!" "The work of passive scientists, poets, and artists is a great support for despotism against those who fight for freedom..."*.

The sanctions imposed by foreign governments and institutions on the Russian scientific sector are not enough. Some journals accept articles with Russian

affiliations. An example is an article in The Astrophysical Journal, published by the *American Institute of Physics*, co-authored by Viktor Sadovnichy, rector of Moscow State University (one of the key and leading Russian universities), who signed the aforementioned letter of rectors in support of the war and is an open supporter of russia's expansionist policy.

Special attention should be paid to grants and programs to support Ukrainian, Russian and Belarusian scientists altogether. These are usually scholarships or employment programs in the EU or the USA. Experts give different assessments of these grants. One point of view of our respondents is that this is how Russia's scientific potential is being drained. Perhaps this will happen in the long run, although it is doubtful that such programs will cause significant damage to Russian science. On the other hand, such grants are morally questionable, as they equalize representatives of the aggressor country (Russia) and the victim country (Ukraine). The conditions for each of these groups are radically different and unequal. Ukrainian male scientists aged 18–60 cannot leave the country due to martial law, and many female scientists do not want to leave their families and leave either, while Russians have no obstacles to leaving the country and make up a significant (or even larger) share of all applicants. Western universities employ many scientists of Russian descent, and it is easier for Russian scientists to find necessary contacts and go to foreign universities than for Ukrainians, if they can even do it. As a result, western universities are filled with people from Russia even more. Because their views on the war and Russian colonialism towards Ukraine cannot be verified, the work of such people in the same group or department as Ukrainians can lead to conflicts and moral suffering for the latter. Another point of view of our respondents is that this way the scientific potential of Russia is being drained. The negative aspects of such grants described above outweigh the theoretical positive ones.

It's time to unleash the expert potential of scientists in the development of civil society, as *Sebastian Dahle*, President of Eurodoc (the European Council of Doctoral Candidates and Junior Researchers), noted on his Facebook page. After 24 February 2022, Eurodoc immediately expressed its solidarity with the people of Ukraine and completely condemned Russia's aggressive war, and created a special working group to support its Ukrainian colleagues. The organization cooperates with key stakeholders in the EU and beyond, and therefore has a possibility to promote concrete measures to support Ukraine.

Conclusions

In a crisis, scientists are an insufficiently visible social group, and the roots of this problem are: limited public access to information about the results of scientific activities, limited direct contacts with business; unsystematic informing of the public, random exchange of information about the needs of science (needs of scientists for research; lack of communication of Ukrainian science abroad).

A desk study on advocacy mechanisms, analysis of expert interviews with stakeholders, monitoring of social media on scientific topics, and an online survey on the needs of Ukrainian scientists led to the following conclusions.

- During the crisis, scientists act in conditions of uncertainty, often take responsibility and make decisions faster than representatives of the relevant executive departments on the ground, act not only to preserve science but also to save lives; storytelling about this can be the basis for positioning science and scientists for public support.
- Advocacy during the crisis concerned the preservation of science in Ukraine. There is a lack of awareness among scientists about institutional mechanisms for improving the situation in the workplace. Unfortunately, scientists have almost no effective advocacy skills, and the ways to protect and promote the interests of the scientific community during the war are even more limited.
- Understanding the needs of a scientist according to different social characteristics opens up different opportunities for interaction with the public: for women scientists, it is about keeping their jobs and creating communication platforms to help with everyday issues; for men scientists, it is about introducing and implementing academic mobility mechanisms; for elderly scientists, it is about inclusion mechanisms and transferring their research experience to the younger generation.
- Advocacy relies mainly on personal communication of the scientist, and the most effective in communication were the institutions that already had established teamwork, a system of connections and were familiar with the methods of working in a crisis.

This White Book allows us to see that scientists are part of civil society, they carry out advocacy actions individually or collectively. The exchange of stories and practices of interaction with the community becomes the basis for creating a good platform for developing sustainable relationships and ensuring support from a wide audience. In solving certain problems, scientists gather around them a team/community that acts as a collective entity based on common values of science preservation.

The study revealed the situation where scientists are unaware of **institutional mechanisms for improving the situation in the workplace.** They perceive the organizational structure for granted, not as something that can be improved based on scientific needs. The bureaucratization of institutional processes makes scientists be creative and solve problems personally, and this does not lead to creation of sustainable models for the preservation and development of science that would be passed down from generation to generation. The system works by itself, and scientists are left alone. To overcome this logistical barrier, we can recommend the introduction of framework programs for interdisciplinary projects in small groups (smart incubators, awareness-raising events, open discussions, filming of popular science content, thematic events in the format of edutainment, etc.). To harmonize the relationship between science and education, it can be recommended to conduct educational trainings for teachers, lecturers, and trainers on science communication for scientists and educators. Education should be a mediator of science, not its customer. The conclusions and recommendations of this book may be useful in formulating the goals of advocacy for the preservation, restoration and development of science.

Understanding the needs of women scientists according to different social characteristics opens up the following opportunities for interaction with the public:

- Advocacy for the needs of women scientists can be aimed at preserving jobs and creating communication platforms in accordance with the needs of general population — children, the elderly.
- Advocacy for the needs of men scientists calls for a review of the specifics of the implementation and enforcement of academic mobility mechanisms, which will also be useful for female scientists, young people and regional and provincial institutions.
- Advocacy for the needs of older scientists coincides with the interests of other older people and people with disabilities, so inclusion mechanisms can be used for this area of advocacy.

These areas are important for understanding the interaction with other stakeholders. For example, the government is obliged to get involved in meeting social needs, businesses benefit from this because of their interest in promoting their unique value propositions and building loyalty of target audiences, and NGOs can offer cooperation in implementing social, humanitarian and cultural projects.

Recommendations

Systemic changes will take place by creating conditions for effective advocacy and promotion of science and scientists through formation of relevant public and non-governmental institutions, restructuring of existing communication platforms, and professional development of science activists. Science requires collective advocacy and, accordingly, work with stakeholders, i.e., government, business, civil society, and international partners.

Laws define the rights of scientists and researchers, but it is not easy to exercise these rights. For the most part, scientists are not aware of the tools to protect their rights and do not always recognize needs as problems to be solved. There is a need to create outreach or educational programs to raise scientists' awareness on this topic.

Cross-sectoral partnership, social cohesion, and creation of communication platforms facilitate effective interaction. A well-established model of cooperation between business and scientists in the frame of grant funding, where scientists can be seen as customers, can become a platform for advocating for preservation, development, and popularization of science at various levels. Due to the constant changes in the context, effective advocacy requires regular monitoring of the mechanisms of ensuring scientific activity, and development of communication of science with stakeholders and civil society.

Based on the obtained data, a number of **recommendations** were developed.

For scientists involved in science advocacy

1. **Diversification of scientific agenda** should be advocated to the media and media influencers. It is necessary to formulate principles of science communication and create a code of ethics for science communication. It is necessary to study the possibilities of social media related to advocacy campaigns, form the informational ground and advocacy goals, and involve influencers who are trusted by both scientists and society. The available media resources in higher education institutions (newspapers, radio, promotional platforms, etc.) should be monitored to find out their ability to publish materials about science in a clear language.

2. In **desk research**, the models and formats that facilitate interaction of science with stakeholders should be analyzed and outscaled to inform about science in interaction with journalists and other media representatives. The discussions with leading media and bloggers working for the general public should be arranged. In particular, there is a need for translation of scientific and applied literature on science advocacy.
3. **International advocacy** will help to increase the level of subjectivity of Ukrainian science and give foreign partners a clearer understanding of the current situation in Ukrainian science and its needs. The best way to advocate your interests is to meet international stakeholders in person. We recommend interested Ukrainian scientists and organizations to actively establish contacts with foreigners (through mutual acquaintances, scientific conferences or schools, participating in them online if it is impossible to travel, by attracting foreign scientists to scientific schools in Ukraine, participating in working groups with foreigners). Knowledge of English is an important condition for effective advocacy, so it is necessary to create conditions for learning it.
4. The level of self-awareness, organization and preparedness of the scientific community should increase. Through **networking**, i.e., creation of formal public associations and informal networks in the field of science, the intellectual potential of advocacy should be strengthened which will have a positive impact on the quality of submitted proposals. Public activity of scientists in social media should be combined with activity during public discussions which will increase the effectiveness of advocacy. As the tentative calendar of public discussions of the Ministry of Education and Science is approved and published before the beginning of each year, stakeholders can plan their advocacy activities for the year ahead.
5. The scientists who want positive changes in the development of Ukraine's scientific field should try to realize their potential in the **public sector**. In this case, according to the legislation of Ukraine, civil service can be combined with scientific activity.

For authorities and public institutions

1. The authorities should be more proactive in informing the scientific community about **public discussions** and explain the importance of a particular regulatory act. An elaborate communication system should be in place to attract more participants in the discussions and timely inform about events. To do this, it is necessary to use social networks and social media, not only official channels, but also to send newsletters to thematic groups and forums. Another way would be to create a Google group for public discussions with the possibility of open access for interested parties. It would be useful to create a database of contacts of experts who participated in previous public discussions and send them invitations to participate in the discussions. Moreover, the list of possible formats for public discussions should be expanded (with account of safety requirements) which is currently limited to digital consultations.
2. The practice of **discussing science development strategies** should be improved. First, it is necessary to develop and test a methodology for conducting strategic discussions with an emphasis on synchronous communication (online and offline) in small groups with the participation of a moderator/facilitator. Second, strategic events should be held regularly and on a more systematic basis, and the scientific community should be involved in discussing the conceptual foundations of public policy at the initial stage of development. The principles of Science for Policy should be implemented in the scientific field. Public hearings should not be formal, but should serve to conceptualize, actualize or problematize

the needs, and can be held both on the initiative of internal stakeholders and international organizations, in cooperation with communities and NGOs. Thirdly, to organize advanced training courses for scientists on science advocacy and civil participation in lawmaking. These courses should be based on real drafts of legal acts that are currently being submitted or planned for discussion. To conduct the courses in the context of limited public finances, funds from international partners should be attracted.

3. The role of **the National Council for Science and Technology Development** in shaping state policy in the field of science should be strengthened given its importance as a key element of the science advocacy mechanism. For this purpose, it is important to implement the provision of the Law of Ukraine "On Scientific and Scientific and Technological Activities", according to which the National Council should meet at least four times a year. Given the specifics of martial law, which may make it impossible for the Prime Minister to attend its meetings as the head of the National Council, the meetings may be chaired by the chairmen of the Scientific and Administrative Committees in turn. The recommendation to hold meetings of the Administrative Committee on the eve of each meeting of the National Council, i.e., at least four times a year, is important. In order to improve the qualifications of newly elected members of the Scientific Committee who do not have experience in lawmaking, the Cabinet of Ministers, with the involvement of specialists from the Parliament (upon agreement), may develop and conduct intensive and long-term training courses on the specifics of lawmaking according to the legislation of Ukraine.
4. The existing **advisory bodies** should be further developed and, in cooperation with other NGOs, create conditions for establishment of new councils representing different target groups (councils of women scientists, council of regional scientists, council of basic sciences, etc.).
5. In order to facilitate more effective science advocacy, the list of **open data** on science mandatory for publication should be expanded. This will allow establishing more meaningful and constructive communication on the development of science.
6. **The National Academy of Sciences** should actively and systematically cooperate with the public sector on a partnership basis, including volunteer initiatives, charitable foundations, and NGOs, develop joint solutions, jointly implement projects to meet the needs of science, and conduct advocacy campaigns.

For public sector

1. **In order to strengthen the intellectual potential** of advocacy, it is necessary to develop workshops and trainings, both informational and specifically advocacy. Projects to build advocacy teams should be developed. To increase the culture of discussion, cross-sectoral events, trainings on facilitation, netiquette, and media communications should be organized. Cooperation with stakeholders can begin with joint trainings on science advocacy (with government, business, media) that will include brainstorming and development of project ideas.
2. **Regular public monitoring** of the work of institutions that are called to act in the interests of scientists (National Academy of Science of Ukraine (NASU), National Research Foundation of Ukraine (NRFU), National Council for Science and Technology, trade unions, etc.) should be ensured to determine the effectiveness and transparency of their work, and promote public awareness of the results.
3. Involvement of NGOs as partners in science advocacy, as well as for cooperation with international donors for community development. The projects to build advocacy teams should be developed. **Promoting networking** proves

itself an effective tool for developing science advocacy. It was thanks to networking that during the first year of the full-scale war it became possible to organize processes to help scientists from abroad, provide access to those scientists who need support, etc. Advocacy actions that will facilitate creation of networking spaces by institutions include organizing special events for interinstitutional, cross-sectoral, and cross-disciplinary interaction, supporting networking platforms, websites, and developing centers of communication and interaction. It is important to inform about the experience of implementing changes in other areas of society, applying existing models, but with account of the needs of both general public and a group for whose benefit advocacy is applied.

4. **Translation of foreign research** on science advocacy, recommendations, and manuals into Ukrainian would be helpful in implementing advocacy activities in Ukraine, especially since there are almost no studies on advocacy of Ukrainian science, although there are analytical reports, described advocacy practices in related areas (administration of relocated universities, some achievements of reforms in the field of medicine, social policy, education, examples of local policy changes, in particular in education, culture towards inclusion, lobbying and implementation of legislative changes on human rights protection, gender policy, combating discrimination and violence).

For higher education institutions

1. **The training of specialists who will facilitate science advocacy**, such as science managers, science journalists, science editors, etc. could be a separate area of work. Their responsibilities and competencies will include the ability to develop the content from key messages to narratives for dissemination to different audiences (stories about scientists who saved the world, interesting facts about Ukrainian science that will increase its visibility in the information field), and organize cross-sectoral discussions, direct action campaigns to actualize science in the context of values, motivate people to unite around the values of which science is a part, create spaces for communication about science, i.e., media platforms, platforms, physical spaces (temporary — exhibitions, presentations, and permanent ones) in science parks, libraries, universities, etc.
2. For capacity development of people who advocate for science (mediators, educators, promoters) curricula and lectures on communication, advocacy, and promotion of science should be developed for scientists in cooperation with stakeholders at the master's level; the topics and tasks on science coverage, promotion and advocacy should be integrated into the existing curricula and courses in the humanities at the bachelor level (media communications, journalism, cultural management, event management, project production, etc.), and the programs to improve the skills of existing scientists should be implemented.

For mass media and media influencers

1. **Providing access to information on the results of scientific activities.** Raising public awareness of science through dissemination of balanced, reliable, and accessible content will increase the level of self-awareness, organization, and preparedness of the scientific community. One of the tasks of journalism is to inform people about the world (promptly, reliably, with the necessary completeness, with account of different views, etc.). Journalism, in particular mass media, should develop the space for informing and communicating with science — preparing news, communicating with experts, developing new formats — talk shows, popular science programs, reports from national parks, etc. Cooperation with media editors at both national and local

levels can be effective in covering science in the media discourse. Interaction with regional media by providing them with information about science helps create spaces for communication, preparation of popularization content which will allow for effective advocacy of science to local authorities, administration of scientific institutions and business, etc.

2. The social media monitoring results show that **regions** need to be better informed about their activities, otherwise science advocacy will be focused only on certain regions which will lead to uneven public support for science in the implementation of certain political and economic decisions.
3. An important goal of advocacy is **building trust** in the scientist in society as an active public figure who has weight and can have influence at the political level. To achieve this goal, it is important to show the lives of scientists as people with ordinary needs, not just scientific ones. The interest in historical dates and participation in cultural events, political discussions and even celebrations emphasizes the civic engagement of scientists. Unfortunately, the format of awarding scientists in Ukraine is not yet used as an advocacy tool, like the Nobel Prize, for example. When covering the awards, the name of the prize and name of the nominee are usually reported, the life story of the scientist and their career are described, but not enough attention is paid to the essence of their research. Meanwhile, these are the occasions that could contribute to diversifying the scientific agenda. To do this, it is necessary to discuss the awarding of scientists on the open platforms, in mass media, and develop cooperation with those who are able to use different advocacy formats and can contribute to understanding between those who are in the topic and those outside the "bubble".
4. There is a need to **reposition scientists** and science in the public space, in particular in mass media. This should be an advocacy goal in the interaction between scientists and mass media and help to build trust of the general public through providing useful, interesting, and important information about science. That is, positioning should be considered as a tool for advocacy, not just popularization.

For international stakeholders

1. To facilitate international advocacy, it is necessary to inform about the opportunities for interaction, **build trust,** and engage scientists in the interaction through motivational programs, which will lead to self-organization in implementation of advocacy projects with the purpose of preserving, developing, and restoring the science.
2. The basic principle of advocacy **"nothing for us without us"** should be applied. Stakeholders should not make decisions on assistance without preliminary communication with Ukrainian experts. Our study has shown that, without this, foreigners have little understanding of the real needs and real situation. For more effective interaction and dialogue, it is necessary to open representative offices of foundations on the ground (Horizon Europe is an example).
3. The key needs are **transparency of criteria**, ease of application, and active involvement of scientists in the search for relevant competitions and programs. This will contribute to positive results in providing necessary resources to the scientific community, preserving and restoring its scientific activities in war conditions.

Development of communities requires organizational resources that should be provided for both at the state level and at the level of building an ecosystem of communities in cooperation with other stakeholders.

References

1. V. Usyk, D. Petryk, O. Malchenko (2021) Advocacy for youth councils. Manual. — Kyiv. with the financial support of the British Embassy in Ukraine and the Active Citizens Program of the British Council in Ukraine. — p. 40 URL: *https://www.britishcouncil.org.ua/sites/default/files/advocacy_for_youth_councils_manual.pdf?fbclid=IwAR0w0nbM_OCIh6WTQJ3JajVhJG3GfY6TJG39YW7IzLsik8Aql187FqDgTvU*.
2. A practical guide to advocacy for civil society organizations (CSOs) and patients' organizations / UNDP in Ukraine — O. Naduta-Skrynnyk — Kyiv, 2022. - p. 107 p. Internet: *www.undp.org.ua* // *www.facebook.com/UNDPUkraine*.
3. Pact (2018) Politically Smart Advocacy: A Guide to Effective Civil Society Advocacy for Sustainable Development. Washington, DC: Pact. Translation: WellDone Translation Agency/ *https://www.pactworld.org/library/politically-smart-advocacy-manual*.
4. Einstein A. (1947). Atomic War or Peace. The Atlantic, 1947. URL: *https://www.theatlantic.com/magazine/archive/1947/11/atomic-war-or-peace/305443/*.
5. Tormos-Aponte Fernando et al. (2023). Pathways for diversifying and enhancing science advocacy. URL: *https://www.science.org* (access: September 11, 2023).
6. Meyer D. S. (2007). Building social movements. In: Creating a Climate for Change: Communicating Climate Change and Facilitating Social Change. L. Dilling, S. C. Moser, Eds. (Cambridge Univ. Press, 2007). Pp. 451–461. URL: *www.cambridge.org/core/books/creating-a-climate-forchange/building-social-movements/FAB1DD4430F143BA230808D0BF5F69F1*.
7. Lutsenko, Anastasiia and Harashchenko, Nataliia and Hladchenko, Lidiia and Korytnikova, Nadezhda and Moskotina, Ruslana and Pravdyva, Oleksandra. The Results of The Survey on The Needs of Ukrainian Scientists (First Wave Report) (January 19, 2023). Max Planck Institute for Innovation & Competition. Research Paper No. 23-03. Available at SSRN: *https://ssrn.com/abstract=4335098*.
8. Lutsenko, Anastasiia and Harashchenko, Nataliia and Hladchenko, Lidiia and Korytnikova, Nadezhda and Moskotina, Ruslana and Pravdyva, Oleksandra. The Results of The Survey on The Needs of Ukrainian Scientists (Part I). URL: *https://kau.org.ua/zvit-nauka-1*.

9. Lutsenko, Anastasiia and Harashchenko, Nataliia and Hladchenko, Lidiia and Korytnikova, Nadezhda and Moskotina, Ruslana and Pravdyva, Oleksandra. The Results of The Survey on The Needs of Ukrainian Scientists (Part II). URL: *https://kau.org.ua/zvit-nauka-2*.
10. Lutsenko, Anastasiia and Harashchenko, Nataliia and Hladchenko, Lidiia and Korytnikova, Nadezhda and Moskotina, Ruslana and Pravdyva, Oleksandra. The Results of The Survey on The Needs of Ukrainian Scientists (Part III). URL: *https://kau.org.ua/zvit-nauka-3*.

Role of society in science advocacy:

1. Mooney, C. (2005). The Republican War on Science. Basic Books: New York, USA. 342 p.
2. Nisbet MC, Scheufele DA. What's next for science communication? Promising directions and lingering distractions. Am J Bot. 2009 Oct; 96(10):1767–78. doi: 10.3732/ajb.0900041. Epub 2009 Sep 3. PMID: 21622297.

Public policy and science advocacy:

1. Pielke Jr, R. A. (2007) The honest broker: Making sense of science in policy and politics. Cambridge: Cambridge University Press, 2007. Pp. ix, 188. URL: *https://doi.org/10.1017/CBO9780511818110*.
2. Cash, D. W., et al. (2003). Knowledge systems for sustainable development. PNAS, 100(14), 8086-91. doi: 10.1073/pnas.1231332100. Epub 2003 May 30. PMID: 12777623; PMCID: PMC166186.

Science communication and advocacy:

1. National Academy of Sciences, National Academy of Engineering (US) and Institute of Medicine (US) Committee on Science, Engineering, and Public Policy. (2009). On Being a Scientist: A Guide to Responsible Conduct in Research: Third Edition. Washington (DC): National Academies Press (US); 2009. PMID: 25009901. Bookshelf ID: NBK214568 DOI: 10.17226/12192.
2. Bubela, T., et al. (2009). Science communication reconsidered. Nature Biotechnology, 27(6), 514–518. doi: 10.1038/nbt0609-514. PMID: 19513051.
3. van Dijk, T. A. (1996). Discourse, power and access. In: Texts and Practices: Readings in Critical Discourse Analysis, eds. C.R. Caldas-Coulthard and M. Coulthard, 84–104. London: Routledge.

Science education and advocacy:

1. National Academies of Sciences, Engineering, and Medicine (2017). Communicating Science Effectively: A Research Agenda. Washington (DC): National Academies Press (US). 2017 Mar 8. PMID: 28406600. Bookshelf ID: NBK425710. DOI: 10.17226/23674.
2. Public and Scientists' Views on Science and Society (2015). Pew Research Center. URL: *https://www.pewresearch.org/science/2015/01/29/public-and-scientists-views-on-science-and-society/*.

Effective methods of science advocacy:

1. Nisbet, M. C., & Mooney, C. (2007). Science and society. Framing Science. Science. 2007 Apr 6; 316(5821):56. doi: 10.1126/science.1142030. PMID: 17412941.
2. Sander van der Linden, Leiserowitz Anthony, Maibach Edward (2019). The gateway belief model: A large-scale replication. Journal of Environmental Psychology. Volume 62. April 2019. Pages 49–58. URL: *https://doi.org/10.1016/j.jenvp.2019.01.009*.

Science advocacy in the digital era:

1. O'Connor, Cailin & Weatherall, James Owen (2019). The Misinformation Age: How False Beliefs Spread. New Haven, CT, USA: Yale University Press. URL: *https://doi.org/10.2307/j.ctv8jp0hk*.

Social and cultural aspects of science advocacy:

1. Wynne, B. (1996). "May the sheep safely graze?" A reflexive view of the expert-lay knowledge divide. In: Lash, S., Szerszynski, B., & Wynne, B. (Eds.). Risk, environment, and modernity: Towards a new ecology. London: Sage, 44–83. URL: *https://www.geos.ed.ac.uk/~sallen/jayne/Wynne%201996%20sheep.pdf*.

Ethics and science advocacy:

1. Resnik DB, Elliott KC. The Ethical Challenges of Socially Responsible Science. Account Res. 2016; 23(1):31–46. doi: 10.1080/08989621.2014.1002608. PMID: 26193168; PMCID: PMC4631672.

Science diplomacy:

1. Arnaldi, S. (2016). Science diplomacy. Foundations and practice, EUT Edizioni Università di Trieste. URL: *https://arts.units.it/retrieve/fffa7860-802f-4844-be48-289f21eaa08c/Arnaldi%20ScienceDiplomacy_ENG_rgb%281%29.pdf*.
2. Krasnyak, O., & Ruffini, P. B. (2020). Science diplomacy. Science, 9, 08. DOI: 10.1093/OBO/9780199743292-0277. URL: *https://www.oxfordbibliographies.com/display/document/obo-9780199743292/obo-9780199743292-0277.xml*.
3. Börner, Katy & Milojević, Staša (2019). Science Forecasts: Modeling and Communicating Developments in Science, Technology, and Innovation. In: Wolfgang Glänzel, Henk F. Moed, Ulrich Schmoch & Mike Thelwall (eds.). Springer Handbook of Science and Technology Indicators. Springer Verlag. Pp. 145–157. URL: *https://philpapers.org/rec/BRNSFM*.
4. Santo Fortunato et al. (2018). Science of science. Science, 359, eaao0185. DOI: 10.1126/science.aao0185.
5. Borner K., Nascimento Silva, Filipi & Milojevic, Stasa (2021). Visualizing big science projects. September 2021. Nature Reviews Physics 3(11):1-9. DOI: 10.1038/s42254-021-00374-7.
6. Legrand T, Stone D. (2018). Science diplomacy and transnational governance impact. British Politics. 2018; 13(3):392–408. doi: 10.1057/s41293-018-0082-z. Epub 2018 Mar 7. PMCID: PMC7149144.
7. Kaltofen, C. and Acuto, M. (2018). Science Diplomacy: Introduction to a Boundary Problem. Glob Policy, 9: 8–14. URL: *https://doi.org/10.1111/1758-5899.12621*.
8. Royal Society (2010). New frontiers in science diplomacy: Navigating the changing balance of power. RS Policy document 01/10. URL: *https://www.aaas.org/sites/default/files/New_Frontiers.pdf*.
9. Science Diplomacy. Foundations and practice (2023). Edited by Simone Arnaldi (University of Trieste). EUT Edizioni Università di Trieste: 154. URL: *https://www.openstarts.units.it/server/api/core/bitstreams/71b32984-3958-4286-b5cb-a26cc28060f3/content*.
10. Flink T. (2021). Why science diplomacy needs evaluative backing. Policy brief. February 2021. Using science for/in diplomacy for addressing global challenges. URL: *https://www.s4d4c.eu/wp-content/uploads/2021/02/WhySDNeedsEvaluativeBacking_PolicyBrief_S4D4C.pdf*.
11. Flink, Tim, & David Kaldewey (2018). The New Production of Legitimacy: STI Policy Discourses Beyond the Contract Metaphor. Research Policy, 47:14–22.
12. Mabe, M., & Ashley, E. A. (Eds.) (2017). The Developing Role of Public Libraries in Emergency Management: Emerging Research and Opportunities.

13. Echeverría, L., Aquino, K., & Widmaier, C. (2020). Science diplomacy and sustainable development goals: a latin american perspective. Science Diplomacy Review, 2(1), 3–13.
14. Müller, C. N. W. (2020). Role of Science Diplomacy in the Framework of Multilateralism: Case of Uruguay's. Accession to Trade Facilitation Agreement. Southern Perspectives on Science Diplomacy, 147–151. URL: *https://www.academia.edu/43023946/Role_of_Science_Diplomacy_in_the_Framework_of_Multilateralism_Case_of_Uruguay_s_Accession_to_Trade_Facilitation_Agreement*.
15. Flink T., Schreiterer U. (2010). Science diplomacy at the intersection of S&T policies and foreign affairs: toward a typology of national approaches. Science and Public Policy, Volume 37, Issue 9, November 2010. Pages 665–677. URL: *https://doi.org/10.3152/030234210X12778118264530*.
16. Moreno Ana Elorza et al. (2017). Spanish Science Diplomacy: A Global and Collaborative BottomUp Approach. Science & Diplomacy, Vol. 6, No. 1 (March 2017). URL: *http://www.sciencediplomacy.org/article/2017/spanish-science-diplomacy-global-andcollaborative-bottom-approach* (*https://www.sciencediplomacy.org/sites/default/files/spanish_science_diplomacy_science__diplomacy.pdf*).
17. Ronda-Pupo G.A. (2021). Cuba-U.S. scientific collaboration: Beyond the embargo. PLoS One. 2021 Jul 22;16(7):e0255106. doi: 10.1371/journal.pone.0255106. PMID: 34293035; PMCID: PMC8297818.
18. Li, X. (2023). Science diplomacy in China: Past, present and future. Cultures of Science, 6(2), 170–185. URL: *https://doi.org/10.1177/20966083231183473*.
19. Serafim, M., Bonilla, K., Bámaca-López, E., Castaneda, A., eds. (2021). Science Diplomacy and Sustainable Development: Perspectives from Latin America. Lausanne: Frontiers Media SA. doi: 10.3389/978-2-88971-535-0.
20. Scientific Diplomacy (2013). France Diplomacy. French Foreign Policy. URL: *https://www.diplomatie.gouv.fr/en/french-foreign-policy/scientific-diplomacy/*.

Annexes

Annexes 1. Summary table of expert interview participants

Code	Name	Position (as of the date of interview)	Organization	Stakeholder*	Region	Interview date
NAUK01	Vyacheslav Zakharenko	Director	Institute of Radio Astronomy of the National Academy of Sciences of Ukraine	Scientific community	Kharkiv	27.05. 2023
NAUK02	Yuri Logvinov	Acting Director	O.Ya. Usikov Institute of Radio Physics and Electronics of the National Academy of Sciences of Ukraine	Scientific community	Kharkiv	26.05. 2023
NAUK03	Serhii Laiievskyi	Director (now — Deputy Director)	V. V. Tarnovsky Chernihiv Historical Museum of	Scientific community	Chernihiv	31.05. 2023
NAUK04	Yuriy Danko	Vice-Rector for scientific and international activities	Sumy National Agrarian University	Scientific community	Sumy	
NAUK05	Tetiana Skarboviichuk	engineer of I category	Paleomagnetic laboratory of S. I Subbotin Institute of Geophysics of the National Academy of Sciences of Ukraine	Scientific community	Demydiv, Kyiv region	19.06.2023

Code	Name	Position (as of the date of interview)	Organization	Stakehol-der*	Region	Interview date
NAUK06	Ivan Moysiyenko	Head of Botany Department	Kherson State University	Scientific community	Kherson	02.06.2023
NAUK07	Olena Khadzhyniva	Acting Rector, PhD in Economy, Professor	Pryazovskyi State Technical University	Scientific community	Mariupol, university relocated to Dnipro	15.06.2023
NAUK08	Oksana Shestakova	Head of Department of Social and Cultural Activities	Oles Honchar Kherson Regional Universal Scientific Library	Scientific community	Kherson	21.06.2023
NAUK09	Alexander Kordyuk	Director	Kyiv Academic University	Scientific community	Kyiv	08.08.2023
NAUK10	Anatoly Zagorodnyi	President	National Academy of Sciences of Ukraine	Scientific community	Kyiv	11.09.2023
BUS01	Yuriy Rebets	Scientific Director	Explogen LLC	Business	Lviv	29.06.2023
BUS02	Maria Lemberg	Director	Fedir Shpyg Charitable Foundation	Business		12.07.2023
BUS03	Nataliia Lopina	Founder	ClinCaseQuest simulation-based training platform	Business	Kharkiv	25.07.2023
CIVIL01	Anton Senenko	Senior Researcher	Institute of Physics National Academy of Sciences of Ukraine	Civil society	Kyiv	24.07.2023
CIVIL02	Yulia Bezvershenko	Co-founder of Science at Risk project		Mediators and science advocates		31.07.2023
CIVIL03	Oleksiy Kolezhuk	Chairman (before October 2023), now — Deputy Chairman	Scientific Committee of the National Council for Science and Technology Development	Mediators and science advocates		07.08.2023
CIVIL04	Yuriy Khalavka	Associate Professor, administrator of Ukrainian Scientists Worldwide group	Yuriy Fedkovych Chernivtsi National University	civil society	Chernivtsi	14.08.2023
STATE01	Denys Kurbatov	Deputy Minister of Education and Science for Science and Innovations	Ministry of Education and Science	State institutions	Kyiv	02.08.2023

Code	Name	Position (as of the date of interview)	Organization	Stakeholder*	Region	Interview date
STATE02	Igor Taranov	Director (before 2023)	Directorate for Science and Innovations, Ministry of Education and Science	State institutions	Kyiv	03.09.2023
MEDIA01	Maryna Synhaivska	Deputy Director General	Ukrainian National News Agency Ukrinform	Media	Kyiv	31.08.2023
INTERNP01	Katherine Younger	PhD, permanent staff member and Research Director	IWM, Ukraine in European Dialogue Program	International stakeholders	Wien	20.07.2023
INTERNP02	Yuri Gogotsi	Director	Drexel Nanomaterials Institute	International stakeholders	Philadelphia	17.08.2023
INTERNP03	Ilona Demchenko	Project Manager	House of Europe	International partners	Kyiv	27.07.2023
INTERNP04	Maud Fichet	Policy officer	European Commission, DG RTD	International partners	Brussels	05.10.2023
INTERNP05	Jan Marco Müller	Coordinator, Science Diplomacy and Multilateral Relations	European Commission	International partners	Brussels	05.10.2023
INTERNP06	Peter McGrath	Coordinator of Science Diplomacy Program	The World Academy of Sciences (TWAS)	International partners	Trieste	03.10.2023
INTERNP07	Cathy Campbell	Science diplomacy	Chair, NASEM RESET Ukraine Action Group. Honorary Fellow, American Association for the Advancement of Science (AAAS)	International partners	Washington	26.09.2023

Annexes 2. Discourse on science in social media. First year of war

Report on the results of social media monitoring. PREPARED BY: OLENA ZINENKO, NADIIIA KORYTNIKOVA, BOHDAN HRUSHETSKYI, OLEH FEIA, SVITLANA GALATA. 30.09.2023

Youscan functionality programmed an automatic text parsing (posts and comments in social media) which allowed collecting a corpus of messages that correspond to the predefined tag cloud. To form a sample from the existing corpus of 1.2 million messages, 4,000 messages were randomly selected, of which 1,300 relevant messages were selected and analyzed (i.e., those messages that were actually related to scientific topics and scientific work). Based on the data obtained, the discourse of Ukrainian science during the war in online media and social networks was analyzed (February 2022 —June 2023).

Annexes 3. Case portfolio.

Case portfolio.
Collection of illustrative scenarios, shared by experts in the interview.
Prepared by: Olena Zinenko, Nadiya Koritnikova, Bohdan Hrushetskyi, Oleg Feia, Svitlana Halata
September 30, 2023

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